Table 6.1. Table showing the cuts applied by the standard preselection for $H \to \gamma\gamma$.

Region	R_9	H/E	idMVA	$\sigma_{i\eta i\eta}$	$\mathscr{I}_{Pho}$ (GeV)	$\mathscr{I}_{Track}$ (GeV)
Barrel	[0.5,0.85]	< 0.08	> 0.9	< 0.015	< 4.0	< 4.0
	> 0.85	< 0.08	> 0.9	–	–	–
Endcap	[0.5,0.9]	< 0.08	> 0.9	< 0.035	< 4.0	< 4.0
	> 0.9	< 0.08	> 0.9	–	–	–

shape cuts are only applied on low-R_9 photons. Above $R_9 = 0.85$ in the barrel and $R_9 = 0.90$ in the endcap, no further preselection cuts are applied. For photons below these R_9 thresholds, cuts are applied on the $\sigma_{i\eta i\eta}$, photon isolation, and track isolation. Finally, the electron veto is required for each photon candidate. This veto decision is designed to identify and remove energy signatures associated with a track, which are likely to be electrons.

Once these single photon preselection cuts are applied, all remaining photon canidates are paired together into a collection of diphotons. For each pair, the higher of the two photons must satistfy $p_T > 35GeV/c$. Paired with the baseline requirement of $p_T > 25GeV/c$, these cuts closely mimic the 30 and 22 GeV/c cuts applied at the trigger level. Then, the $p_T/M_{\gamma\gamma}$ cuts are applied to the diphotons. At this point, many of the possible photon combinations have been removed. A final diphoton combination must be selected, however, as HiggsDNA is designed to output a single pair of photons. The remaining pairs in the event are sorted by their diphoton p_T, and the pair with the highest value is written out into the event [34]. These cuts, in their totality, are shown in table 6.1

In addition, there are cuts on the $p_T/M_{\gamma\gamma}$ applied as part of what are termed the Standard Analysis(SA) cuts. These are kept separate from the single photon preselection cuts, as some analyses may want to make adjustments to them. For the Run 3 preselection, a geometric mean $p_T/M_{\gamma\gamma}$ cut is used [16]. The cut on the second (sublead) photon is straightforward, at $p_T^{\gamma 1}/M_{\gamma\gamma} > 1/4$. The second cut, however, uses the geometric mean of the two p_T values as $\sqrt{p_T^{\gamma 1} p_T^{\gamma 2}}/M_{\gamma\gamma} > 1/3$. There is no explicit $M_{\gamma\gamma}$ cut applied within the preselection, as the possibility for high- and low-mass Higgs analysis is left open. For the purposes of comparing this preselection to the new MVA-based selection, only the standard mass Higgs is considered.

So, a loose cut of $90 < M_{\gamma\gamma} < 180 GeV/c^2$ is applied to all resulting diphotons.

6.2 Development of MVA-Based Offline Preselection

Since the offline selection is designed to cover the HLT used, it is clear that applying the older preselection described above on top of the MVA-HLT is a suboptimal strategy. In fact, since these cuts mirror the standard HLT quite closely, it is expected that applying the standard preselection on top of the MVA-HLT would likely reduce or even entirely remove the signal efficiency gains obtained by the new trigger. As a result, an alternative set of preselection cuts must be proposed that allows the trigger-level gains to be realized at the offline analysis level.

Just as the standard preselection cuts are designed to not only cover the standard HLT but work in much the same way, a new preselection can be created with maximum similarity to the MVA-HLT. One major complication of this is that the same BDTs cannot be used for both the trigger and offline selection. Since the reconstruction differs for the NanoAOD vs the RAW, the input variable distributions will be different enough that the models are not directly comperable. Instead, a new BDT-based preselection framework must be developed and its coverage of the MVA-HLT must be analyzed.

6.2.1 Creation of a Single Photon $\gamma + Jet$ BDT

As discussed in section 6.1, there already exists an MVA within the standard preselection. While this idMVA is now a centralized part of the NanoAOD format, historical preselections used an MVA trained of $\gamma + Jet$ MC [34]. Events in $\gamma + Jet$ MCs will contain one real (prompt) photon and one jet that looks like a photon, termed a "fake". This MC provides a very easy-to-use training sample, as one can simply use the prompt γ as the signal photon and the fake as the background photon. This standard $\gamma + Jet$ idMVA is usually only used to make relatively loose cuts as part of the much broader preselection cuts as described above.

The original idMVA was trained using many of the variables from the preselection as input variables. These same input variables were used as a starting point for a new $\gamma + Jet$ MVA,

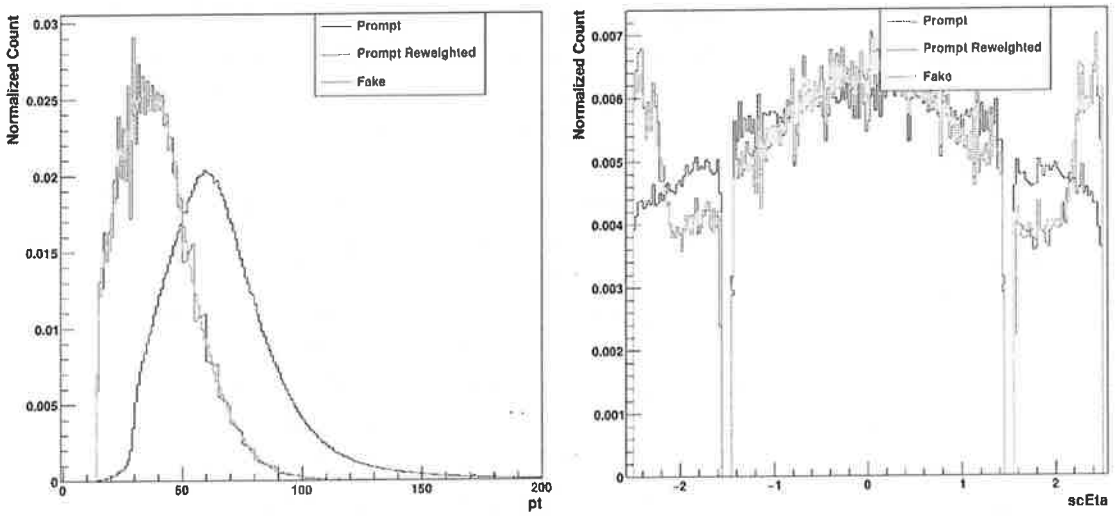
Table 6.2. Table of input variable names for the offline $\gamma+Jet$ MVA, and a short description of each. The final two variables are only used for training the endcap.

Variable Name	Description
Raw Energy	The total energy deposited in the 5x5 supercluster
R_9	The proportion of the energy contained in the central 3x3 crystals relative to the raw energy
S_4	The proportion of the energy contained in the central 2x2 crystals relative to the raw energy
$\sigma_{i\eta i\eta}$	The covariance in terms of crystal spacing (i_η)
$cov(i\eta i\phi)$	The covariance between the η and ϕ coordinates
η_{SC}	The eta coordinate of the center of the supercluster
η Width	The supercluster width in η
ϕ Width	The supercluster width in ϕ
H/E	The ratio of the total supercluster energy measured in the HCal to that measured in the ECal
PhoIso03	The photon isolation
Charge Iso w.r.t. Worst Vertex	The charged isolation for the worst vertex determined by the vertexing algorithm
Charge Iso w.r.t. Chosen Vertex	The charged isolation for the chosen vertex determined by the vertexing algorithm
ρ	The event energy density
esEffSigmaRR	SigmaRR from the preshower detector
esEnergyOverRawE	Ratio of the preshower energy to the total supercluster energy

with the addition of H/E. By adding this variable to the inputs, the new $\gamma+Jet$ MVA contains the same information available to the standard preselection described above. Additionally, this allows the offline selection model to more closely mirror the MVA-HLT. The input variables used are shown in table 6.2. As with the MVA-HLT, the barrel and endcap models are trained separately, with individual photons selected based on their location in the detector.

The $\gamma+$ Jet BDT is designed to be as blind to the p_T and η_{SC} of the photons as possible, so as not to bias the kinematic acceptance; prompt photons tend to be higher in p_T, and it is best to not use this fact as a discriminator if possible. So, then, a process has been developed to reweight the prompt photons to more closely match the p_T and η_{SC} distributions of the fake photons. A 2D distribution of the p_T and η_{SC} is first constructed for both the prompt and fake photons, with relatively coarse binning. For each bin, a factor between the prompt and fake is

found such that a weight can be assigned to each prompt photon. These weights are then stored in the event, and can be applied during the training to each prompt photon in order to avoid the use of the p_T and η_{SC} in discriminating between real and fake photon candidates. Figure 6.1 shows the effects of this reweighting, where the original prompt distribution (black) is reweighted (blue) to closely match the distribution of the fake (red) in both p_T annd η_{SC}.



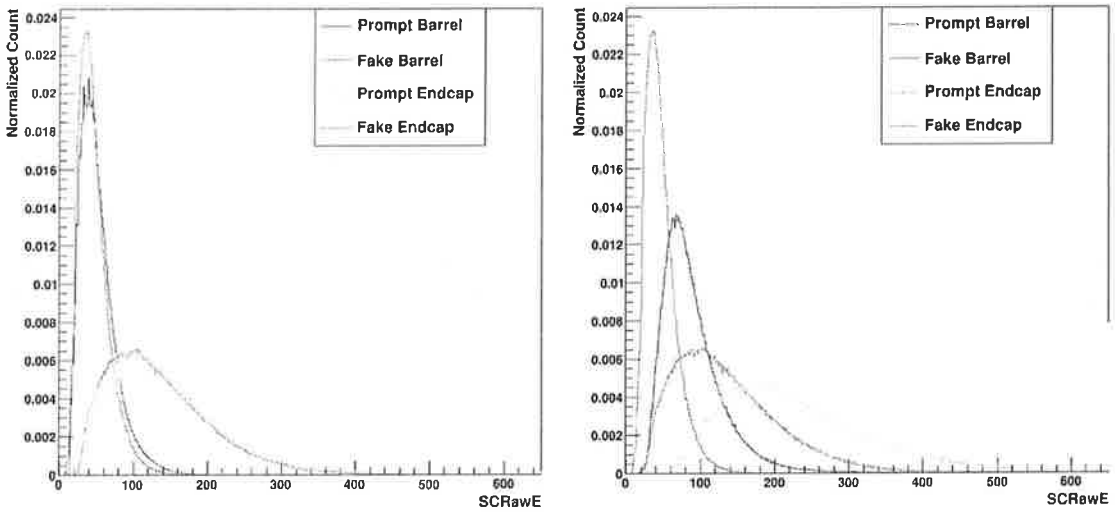
(a) p_T **(b)** η_{SC}

Figure 6.1. Plots of the p_T and η_{SC} reweighting, with the untransformed prompt (black), the reweighted prompt (blue), and fake (red) shown. It can be seen how the reweighting procedure causes the prompt distributions to closely match those of the fake photons.

As with the trigger level BDT, the separation between the signal and background in the various input variables can be examined. For the subsequent plots (6.2-6.8), the prompt and fake will be shown for both the barrel and endcap, as the trainings are separated. The prompt photons will be shown in dark and light blue for the barrel and endcap, respectively. The fake distributions will be shown in red and pink for the barrel and endcap, respectively. All the input variable distributions have been normalized to 1, so that their shapes can be directly compared.

The first two variables, SCRawE and η_{SC}, are not designed to be clear discrimination variables; since the raw energy will be directly correlated with the p_T and η_{SC} weights, the

reweighting procedure ensures that the prompt and fake are relatively similar in this variable. At left, in figure 6.2a, the reweighted distribution can be seen. At right, in figure 6.2b, the unreweighted distributions can be seen. While the reweighting procedure has a large effect on this energy variable, the effect on other variables is relatively minor. It can still be seen, clearly, that the distributions are quite different for the barrel and endcap. Of course, the η_{SC} is also directly reweighted as shown in figure 6.1b. BDTs, however, are very adept at finding multi-dimensional correlations between variables. Providing the supercluster energy and η, then, allows the model to access more information about the event and achieve a more holsitic understanding of how prompt and fake photons look different. While these variables are not directly useful for discriminating between prompt and fake, they are still a vital component of the BDT.



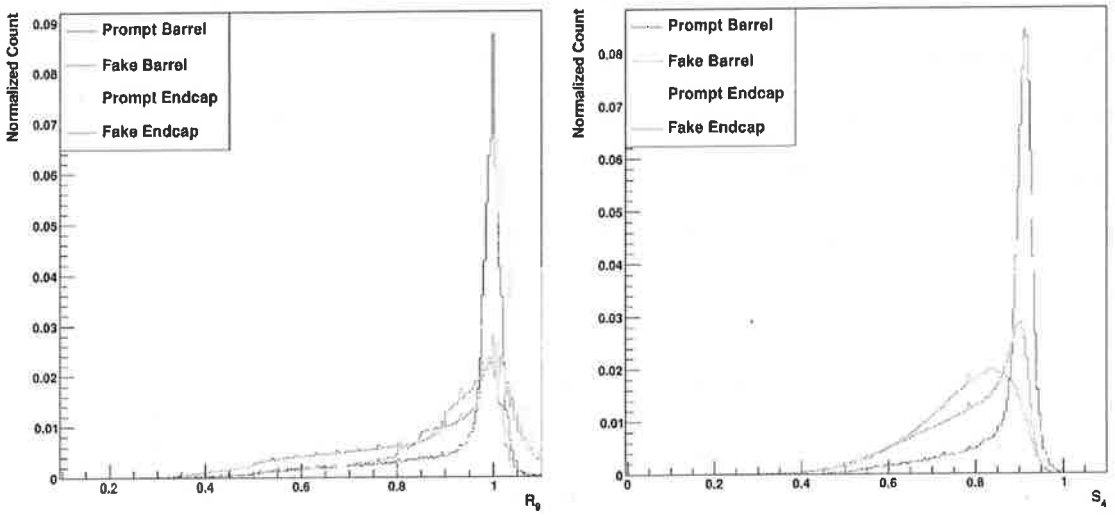
(a) Reweighted Raw Energy

(b) Unreweighted Raw Energy

Figure 6.2. Plots of the Supercluster raw energy, where the left has been reweighted and the right has not. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Next, the energy ratios, R_9 and S_4 can be examined. It is expected that a real (prompt) photon will have relatively well-localized energy; since the photon is a relatively clean signal

in the detector, it is expected that most of the energy will be deposited in the center of the supercluster. It is clear that this is the case for both the barrel and endcap, as seen in figure 6.3. Since the fake photon corresponds to a jet, it is expected that the energy deposition will be more spread in the supercluster, which can also be seen in the plots. As a result, R_9 and S_4 can be relativelty powerful discrimination variables.

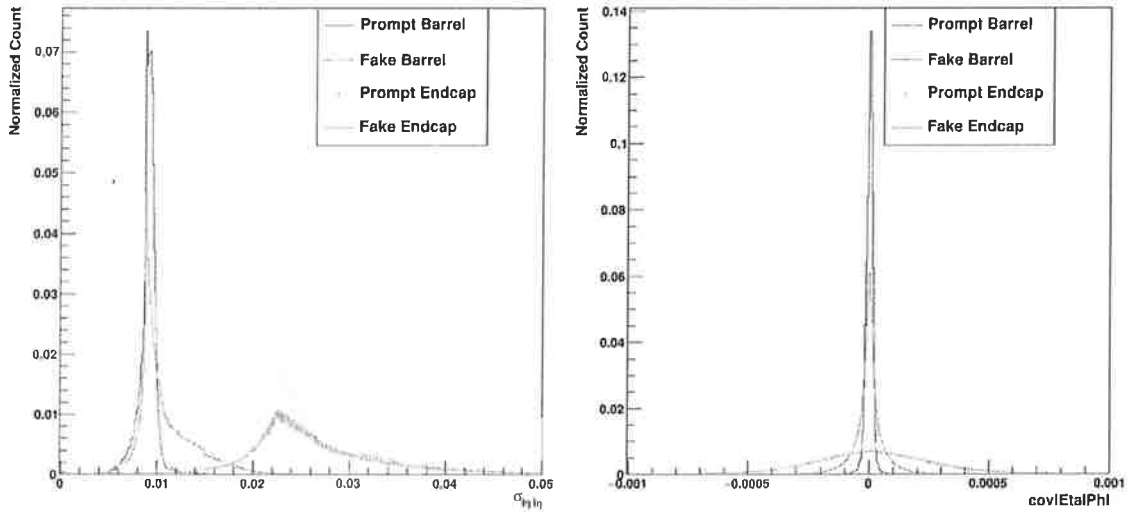


(a) R_9 **(b)** S_4

Figure 6.3. Plots of the R_9 and S_4. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The next shower-shape variables, $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$), provide information about the energy depositions in terms of the crystal spacing. $\sigma_{i\eta i\eta}$ quantifies the standard deviation of the energy deposited in terms of the η crystals in the supercluster. Cov($i\eta i\phi$), on the other hand, quantifies the covariance between the energy deposited in the η and ϕ directions. Since these measurements both depend on the crystal size and spacing, their values differ considerably between the varrel and endcap. This can be seen clearly in figure 6.4. In the barrel, the $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$) both prove to be very good discrimination variables; the fake photons tend to have considerably larger spread in both measures compared to the prompt photons. In the endcap,

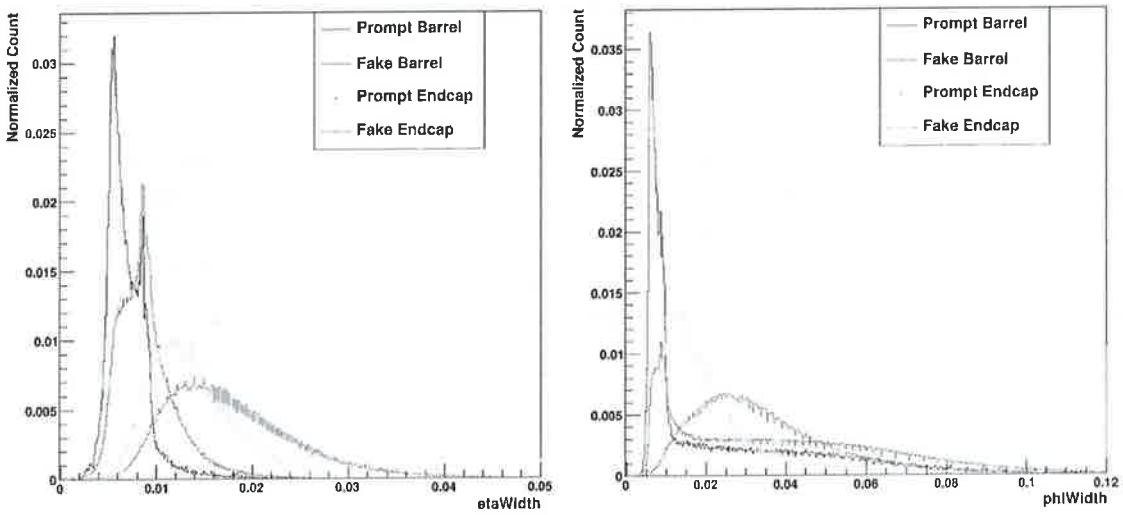
the signal $\sigma_{i\eta i\eta}$ is still clearly well-separated from the background, though the separation in cov($i\eta i\phi$) is much smaller.



(a) $\sigma_{i\eta i\eta}$ **(b)** cov($i\eta i\phi$)

Figure 6.4. Plots of the $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$). The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

Similarly to the $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$), the η and ϕ widths are powerful shower-shape variables that provide some measure of how well-localized the energy deposition is. It is expected that these values are different for the barrel and endcap, as the crystal size and spacing are different in the two different regions of the detector. This can be seen in both plots in figure 6.5. Kinematically, the effects of the magnetic field must be considered here. Since the magnetic field acts in the ϕ direction but not in η, it is expected that photon showers will be spread out more in ϕ than η. This separation can clearly be seen in the endcap for the ϕ width in figure 6.5b, and it can be seen that a larger proportion of the barrel fake photons extend to higher values compared to the prompt photons in the barrel. Decent separation can still be seen for the barrel and endcap in η width in figure 6.5a.



(a) η Width

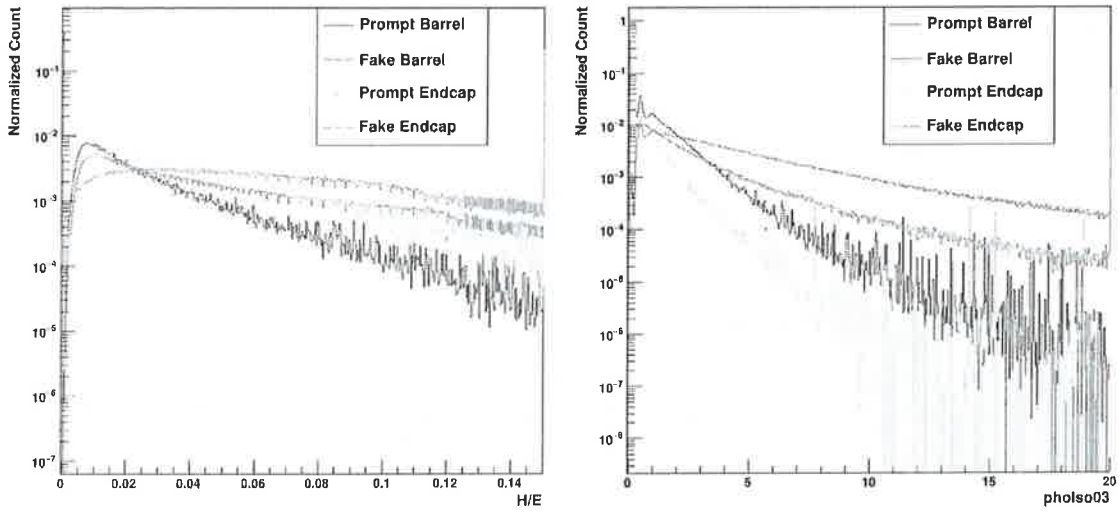
(b) ϕ Width

Figure 6.5. Plots of the η and ϕ widths. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The measure of the HCal energy vs the ECal energy, H/E, provides a powerful measure of whether the energy deposition more closely matches an electromagnetic process – such as a photon or electron – or a hadronic process such as a jet. It is expected that most photons will have a value consistent with 0, as they should deposit all of their energy into the ECal. There is, of course, some level of energy leakage that ensures that even photons often do not have an H/E of identically 0. Clearly, however, the prompt photons in both the barrel and endcap strongly peak near 0 and fall off rather quickly below this. Note that figure 6.6a is a log plot, so that the secondary peak below 0 is down by more than a factor of 10 relative to the primary peak at 0. Since the fake photons are jet signatures designed to mimic a prompt signature, they still exhibit a strong peak near 0. It can be seen, however, that the distribution extends to much higher values of H/E overall.

The photon isolation shown in figure 6.6b is another very strong discrimination variable. This variable is the energy isolation of a signature assuming a particle flow identification as a

photon, and is the sum of energy in the isolation region in GeV. The isolation is taken in a cone with $\Delta R = \sqrt{\eta^2 + \phi^2} = 0.3$ surrounding the supercluster. It is expected that the prompt photons will have a lower photon isolation energy than the fake, which can be seen in the figure. While the prompt photons decay quite quickly with increasing photon isolation, it can be seen that the fake photons have a larger proportion of high isolation energies for both signal and background.

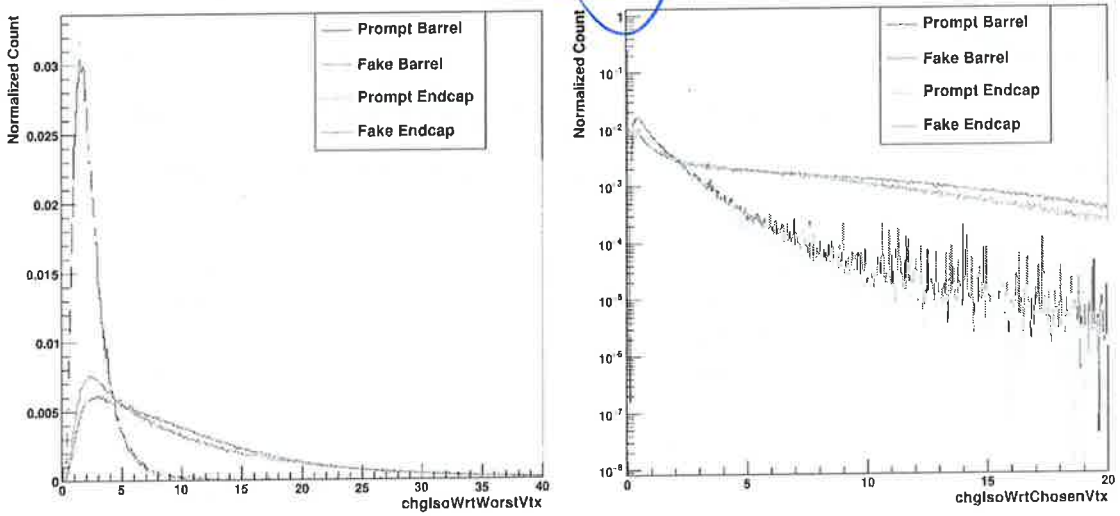


(a) H/E **(b)** Photon Isolation

Figure 6.6. Plots of the H/E and photon isolation. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The charge isolation variables are the next input variables, and represent the isolation energy (in GeV) relative to a specific vertex. The vertexing described in section 3.3.1 provides multiple vertices at the NanoAOD level. Isolations are measured relative to both the worst possible vertex (figure 6.7a), and the final chosen vertex (figure 6.7b). The worst vertex is the choice that provides the highest possible isolation energy, whereas the best vertex is the one chosen by the particle flow algorithm. While the chosen vertex often has 0 isolation energy, the worst vertex very rarely does. In both cases, however, the separation between signal and background is clear for both the barrel and endcap. The charge isolation w.r.t. worst vertex,

specifically, shows very clear and powerful spearation.

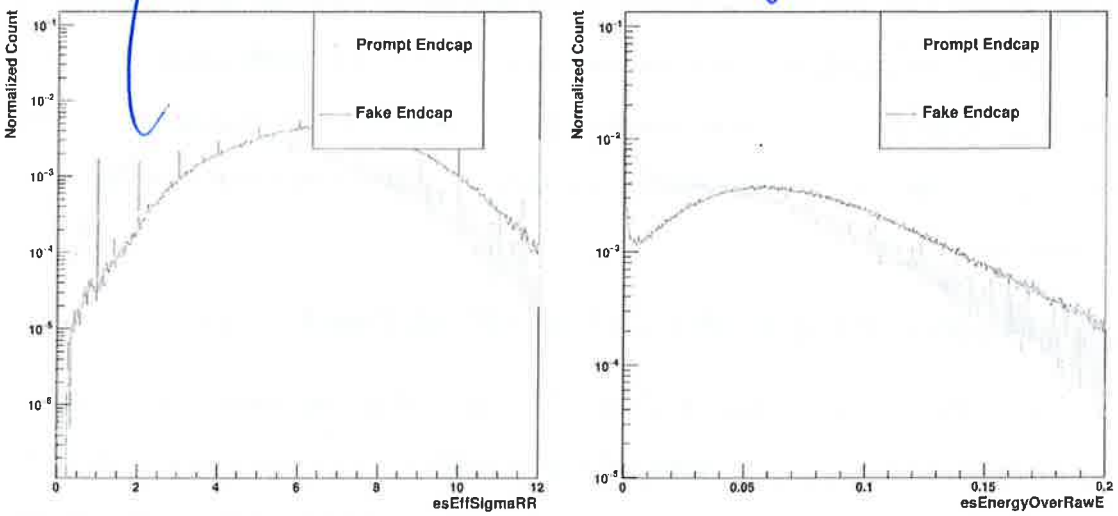


(a) Chg. Iso. W.r.t. Worst Vertex

(b) Chg. Iso. W.r.t. Chosen Vertex

Figure 6.7. Plots of the charged isolations with respect to the chosen and worst vertices. The prompt signal is shown in dark blue for the barrel and light blue for the endcap, whereas the fake background is shown in red for the barrel and pink for the endcap.

The final two variables in the model are only used in the endcap models. In fact, these variables only exist for endcap events. Both of them come from the preshower detector that only exists in front of the endcap; the esEffSigmaRR is a measurement of the simgaRR in the preshower detector, whereas the esEnergyOverRawE is the ratio of the energy deposited in the preshower detector over the total raw supercluster energy of the photon candidate. While some differences can be seen between the prompt and fake distributions, there is not an especially marked separation in these two variables. As a result, it is not expected that they will be particularly powerful discrimination variables on their own, though high-dimensional correlations with other variables may still be leveraged by the BDT.



(a) ES Eff. Sigma RR **(b)** ES Energy over Raw E

Figure 6.8. Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The prompt signal is shown in light blue, whereas the fake background is shown in pink.

Comparing to the MVA-HLT variables shown in 5.1, it can be seen that a few variables have been added at the offline selection level. The photon isolation is analogous to the ECalIso used at the trigger level, but is calculated differently. As discussed in section 3.3.1, the offline reconstruction uses the complex particle flow algorithm to reconstruct the event. So, compared to the ECal isolation at the trigger level, additional corrections and refinements are applied to the offline isolations based on both vertexing and the choice of particle hypothesis. The two charged isolations used are entirely new additions, as the MVA-HLT is calculated before any complex vertexing is implemented. By using both the isolation relative to the chosen and the worst vertices, the offline BDT can be less beholden to the accuracy of the vertex selection; vertexing is a very complex process in the reconstruction algorithm, and serves as a major source of systematic error.

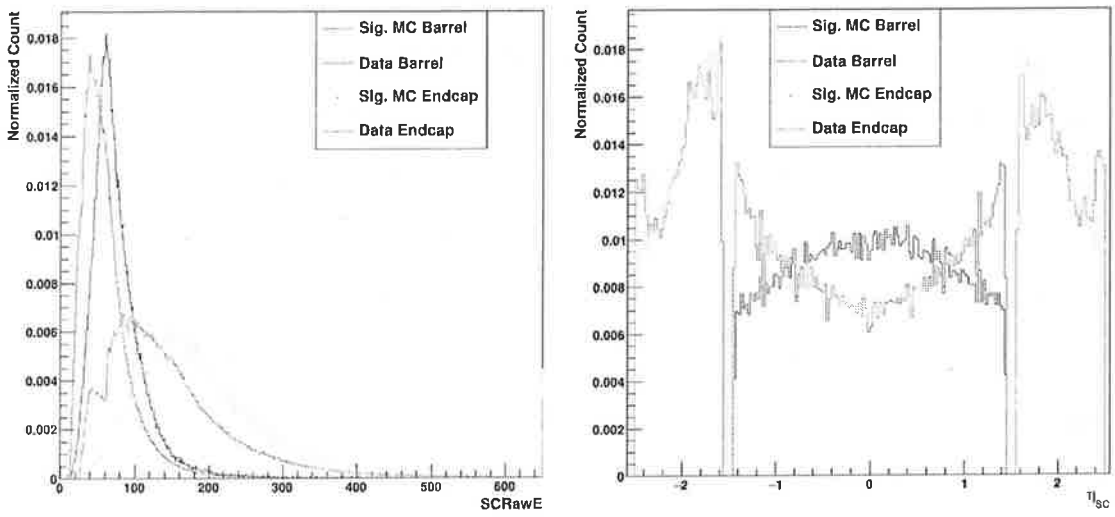
The hyperparameter optimization at the offline level was performed very similarly to the MVA-HLT; the max depth and learning rate were varied in tandem until an optimal point was found. The same process of early stopping within XGBoost was used, as well. The

hyperparameters were, however, found to be quite different from the trigger-level optimization. Since there are considerably more input variables used (13/15 for the barrel/endcap versus the 8 for the trigger), the max depth was necessarily higher. Consequently, the learning rate was also made lower. The optimum hyperparameters were, then, found to be a maximum depth of 18 and a learning rate of 0.06.

6.2.2 Creation of a Single Photon Data/MC BDT for $H \to \gamma\gamma$

While the above $\gamma + Jet$ BDT can help differentiate real photon candidates from fakes, it cannot, in principle, properly differentiate a photon candidate coming from a signal process from a photon candidate coming from a background process. Since the EGamma trigger selections will already cut out many events with jet-like photon candidates, much of the $H \to \gamma\gamma$ preselection is aimed at removing real photons that do not originate from the Higgs boson. There are many sources of real photon backgrounds, including single beam backgrounds arising from the interaction of a single beam with the gas in the detector. Since these are real photons, their isolation and showershape variables alone are often not enough to reject them. For the standard preselection, the p_T and $p_T/M_{\gamma\gamma}$ cuts achieve this necessary rejection.

Furthermore, the $\gamma + Jet$ BDT does not properly mirror the MVA-HLT; since the trigger-level BDT was training using Higgs MC as the signal and real data as the background, the offline selection should employ a similar strategy. So, in the same way that the trigger was trained using signal photon candidates originating from a ggH MC sample and background photon candidates originating from the data. The same input variables used in the $\gamma + Jet$ BDT were then used to train and optimimize the new model. These input variables are plotted in the same way as for the previous section, with the barrel and endcap signal being shown in dark and light blue, respectively. Meanwhile, the barrel and endcap background will be shown in red and pink.



(a) Raw Energy

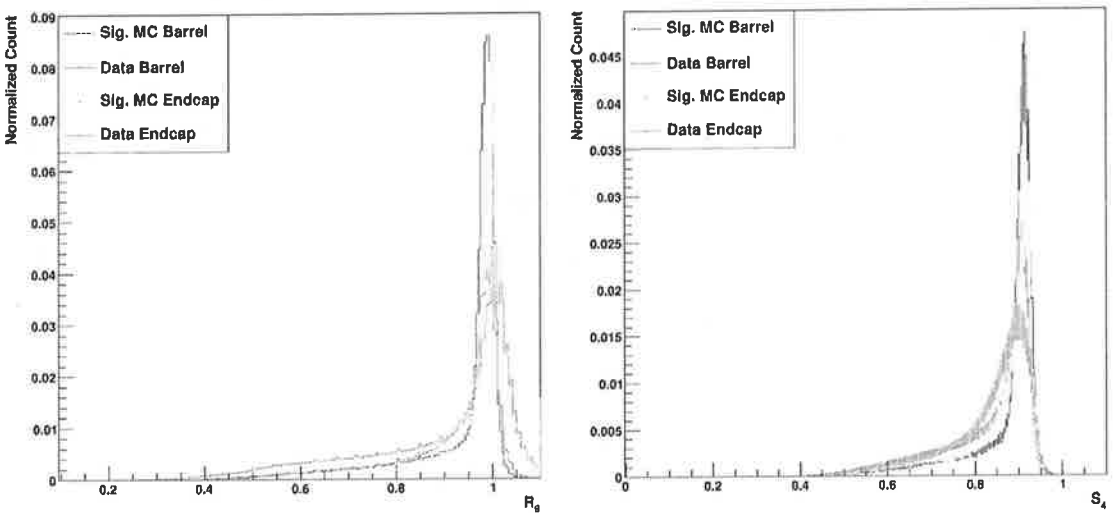
(b) η_{SC}

Figure 6.9. Plots of the Supercluster raw energy and ηSC. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

For the γ + Jet BDT, the process of p_T and η_{SC} reweighting was described. Since the prompt and fake photons originated from the same simulated events, there were the same number of them with correlated kinematics. This meant that the reweighting process was quite straightforward. When using $H \to \gamma\gamma$ MC as signal and data as the background, however, this reweighting procedure makes less sense. Since the number of events is quite different and the distributions in η_{SC} give some real information about the events themselves, the direct reweighting has not been applied here. Instead, it is worth considering the goal of the original reweighting for the γ + Jet MC. When selecting prompt from fake photons, it was vital to not bias the diphoton mass distribution in any way; shaping the mass distribution with a selection means that the analysis is no longer blind to different mass hypotheses, and points to a lowered overall kinematic coverage. As long as the mass distribution is not biased by the chosen selection, the p_T and η_{SC} reweighting is not a vital component of the selection. So, then, the data/MC BDT can be constructed without this reweighting, but special attention must be paid to ensure that the

background mass distribution is not being shaped by this selection.

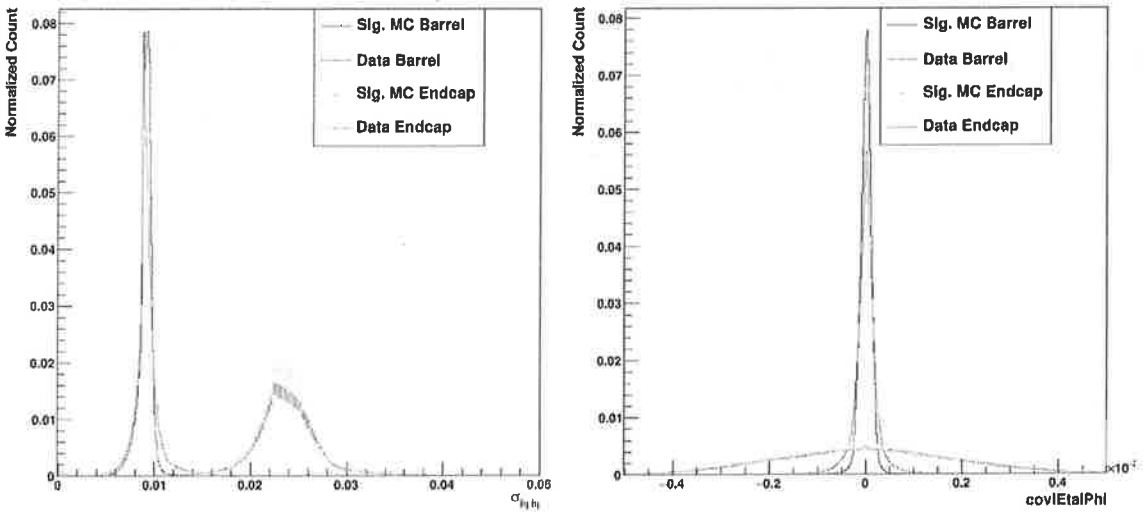
As seen in figure 6.9a, the supercluster raw energy differs a bit between the signal MC and the data. While the difference isn't as pronounced as it was for the unreweighted distributions for the γ + Jet prompt and fake (figure 6.2b), there is still more separation here than in the reweighted version for γ + Jet. While this variable still provides useful information for the BDT to make high-dimension correlations, it must still be verified that this difference doesn't bias the background mass. The features of the η_{SC} distributions for the data and MC differ considerably more than those for the γ + Jet prompt and fake distributions, however. Particularly in the barrel, it can be seen that the signal MC tends to be much more centeral, whereas the data peaks strongly towards to high-η regions of the barrel. The endcap distributions, however, are in decent agreement.



(a) R_9 (b) S_4

Figure 6.10. Plots of the R_9 and S_4. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

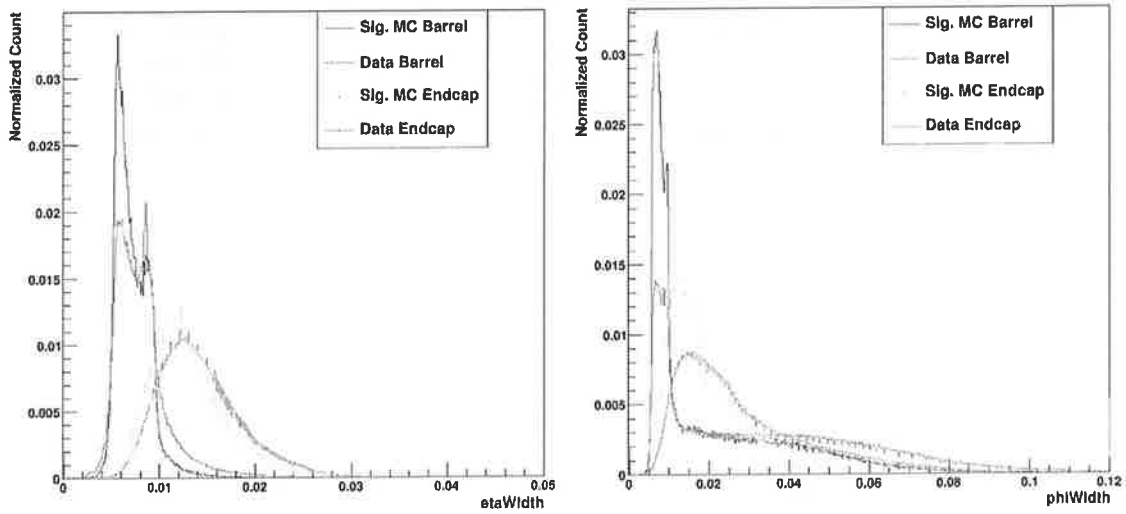
As with the γ + Jet distributions, the R_9 (figure 6.10a) and S_4 (figure 6.10b) variables for the signal tend to be much more strongly peaked near 1 compared to the background.



(a) $\sigma_{i\eta i\eta}$ **(b)** cov($i\eta i\phi$)

Figure 6.11. Plots of the $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$). The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

In figure 6.11, it can be seen that the signal and background distributions are quite similar for both $\sigma_{i\eta i\eta}$ and cov($i\eta i\phi$). Compared to the γ + Jet BDT, this means that these shower-shape variables will be much less powerful. Likely, due to the MVA-HLT selection applied, many of the more jet-like photons have already been removed from this training sample.

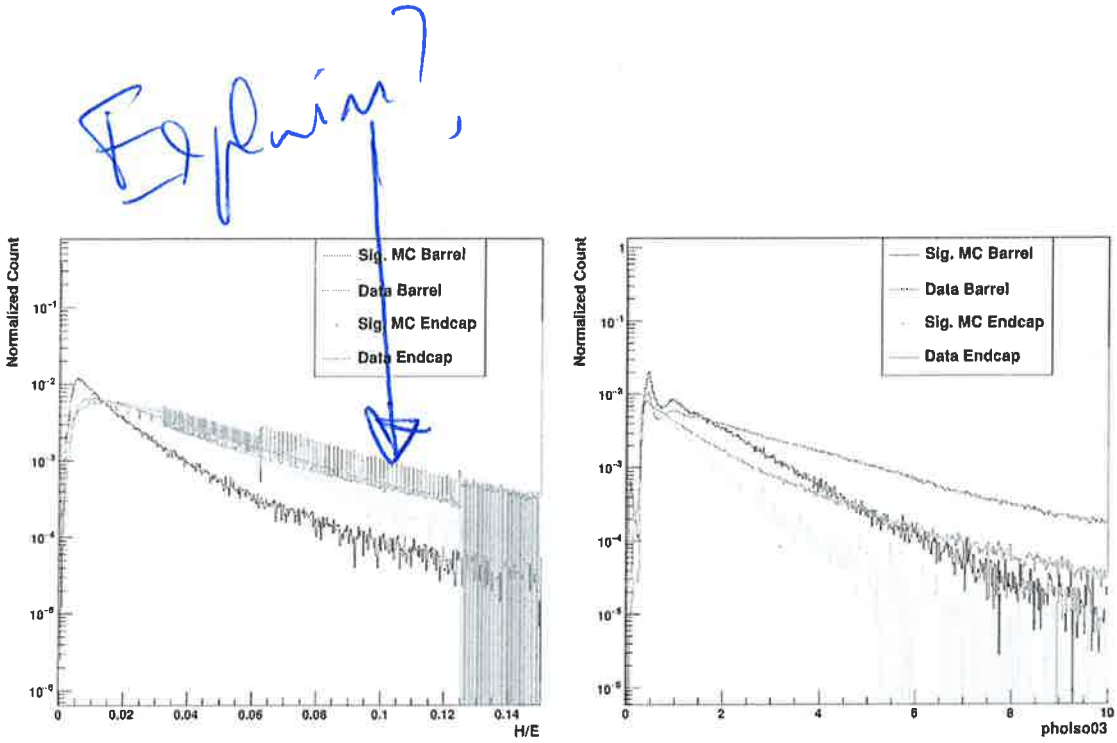


(a) η Width

(b) ϕ Width

Figure 6.12. Plots of the η and ϕ widths. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

The η and ϕ widths in figure 6.12 show similar separation between signal and background relative to the γ + Jet distributions.

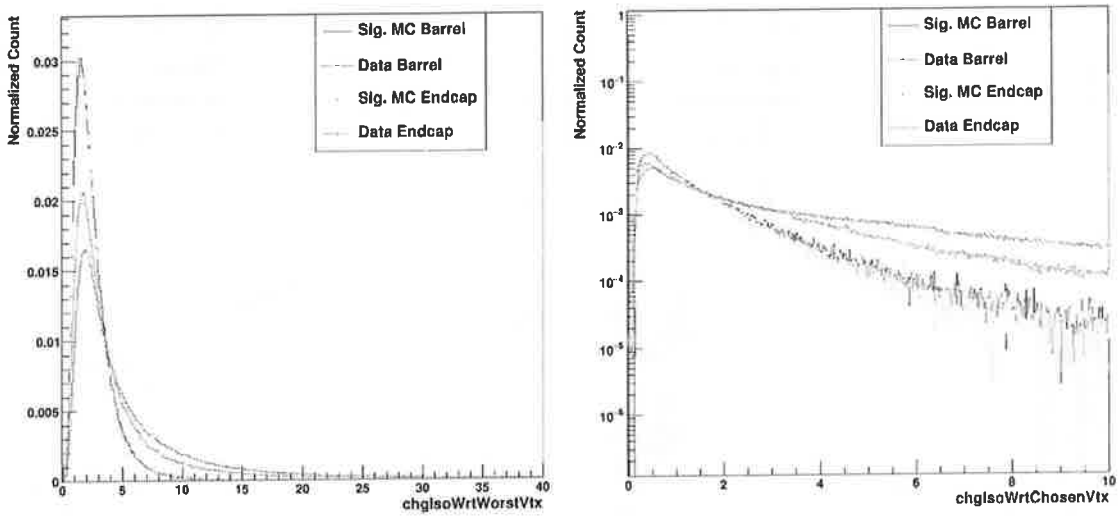


(a) H/E

(b) Photon Isolation

Figure 6.13. Plots of the H/E and photon isolation. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

The H/E and photon isolation in figure 6.13 show decent separation between the signal MC and data in both the barrel and endcap. The barrel and endcap for the data are quite similar for H/E, while the signal MC shows considerably more variation between the two detector elements. Overall, H/E remains a quite powerful discrimination variable in these new samples. Similarly, clear differences can be seen between the signal and background in the photon isolation.

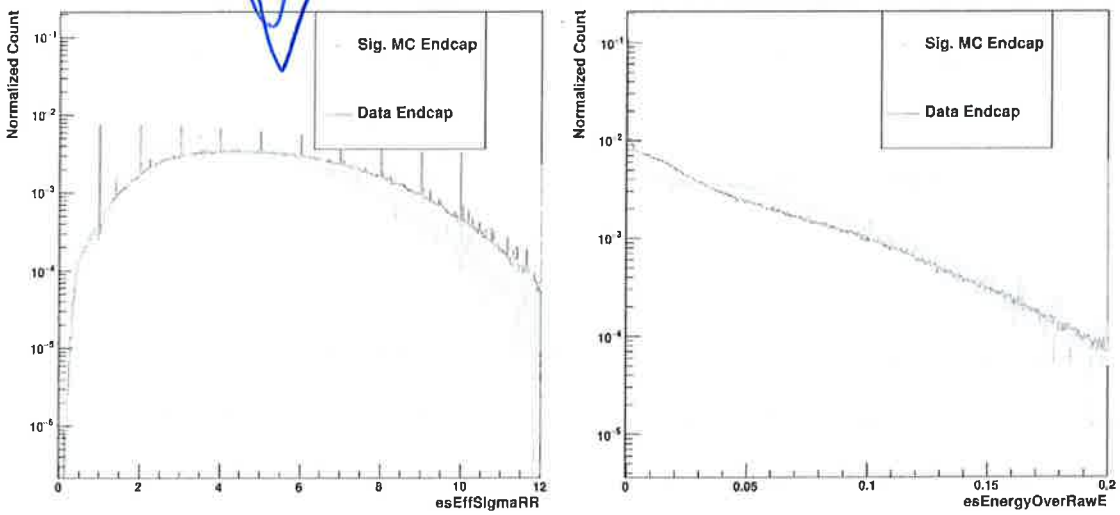


(a) Chg. Iso. W.r.t. Worst Vertex

(b) Chg. Iso. W.r.t. Chosen Vertex

Figure 6.14. Plots of the charged isolations with respect to the chosen and worst vertices. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

In figure 6.14, it can immediately be seen that the two charge isolations are quite similar for the signal and background in both the barrel and endcap. While the values tend to extend a bit higher for the background overall, the shapes are far more similar between the signal and background classes relative to the γ + Jet distributions.



(a) ES Eff. Sigma RR

(b) ES Energy over Raw E

Figure 6.15. Plots of two endcap-specific variables: esEffSigmaRR and esEnergyOverRawE. The signal MC is shown in light blue, whereas the data (background) is shown in pink.

As with the γ + Jet BDT, the two endcap-specific input variables do not show major separation.

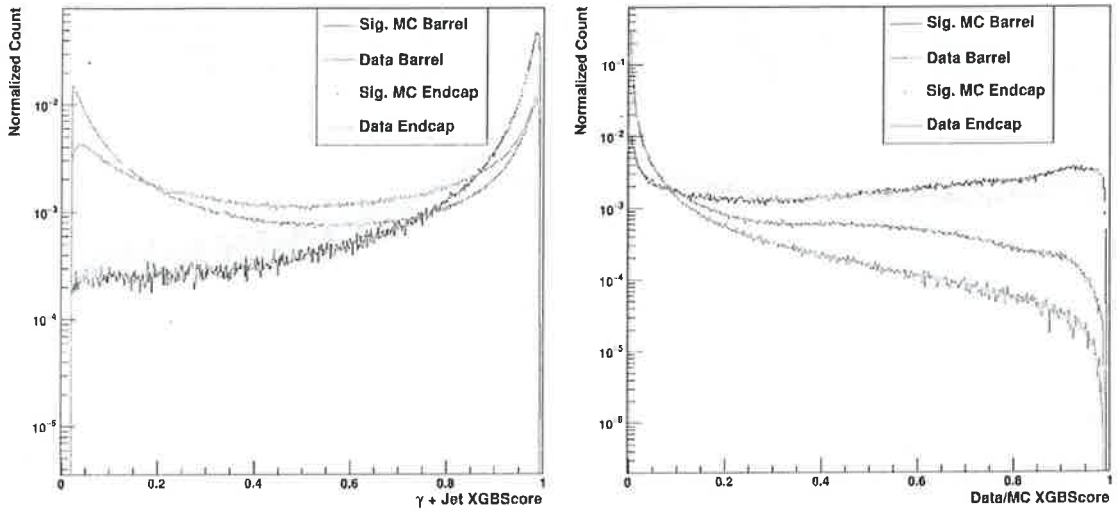
The same process of hyperparameter optimzation as the previous models was used. Notably, the signal MC and data show much more similarity in the input variable distributions relative to the γ + Jet samples. As a result, the model must become a bit more complex to achieve the desired discrimination power. The same experimental optimization process led to a maximum depth of 18 paired with a learning rate of 0.025. While the maximum depth is the same as that of the γ + Jet model, the learning rate is quite a bit lower than the 0.06 used for that BDT.

6.2.3 Determination of Loose Single Photon MVA Cuts for $H \to \gamma\gamma$

The standard preselection first makes decently loose cuts on individual photons, then tighter p_T and $p_T/M_{\gamma\gamma}$ cuts on selected diphoton combinations. A very similar strategy can be used for the BDT-based offline selection. Firstly, loose cuts on the scores of the BDTs developed above can be applied to the individual photon candidates, which will reduce the number of

possible diphoton combinations and cut out many events with no reasonable photon candidates. Then, with the list of individual photons cleaned up significantly a tighter selection can be applied to the resulting diphoton objects for the selection of a final pair of photons.

The single photon BDT scores can be seen in figure 6.16. The γ + Jet scores can be seen at left in figure 6.16a. It is immediately obvious that the separation is quite good at low scores – the data is given very low scores for many of the photons, whereas there is very little signal MC at a score near 0. There is, clearly, still a large quantity of data given a score closer to 1. These likely correspond to real photons in the data, which the γ + Jet BDT is not designed to deal with. As a result, the chosen score cut on the γ + Jet score can be decently tight to remove much of the data without having a large impact on the signal efficiency. The Data/MC scores can be seen at right in figure 6.16b. While there is very little data with a score near 0, the signal distribution is not as well-peaked as that of the γ + Jet. Clearly, however, the data peaks very strongly near 0 – upwards of 20% of the data is located in the first bin. This means that relatively tight cuts on this score will be able to drastically reduce the data acceptance, though the amount of signal MC cut out must be carefully monitored.

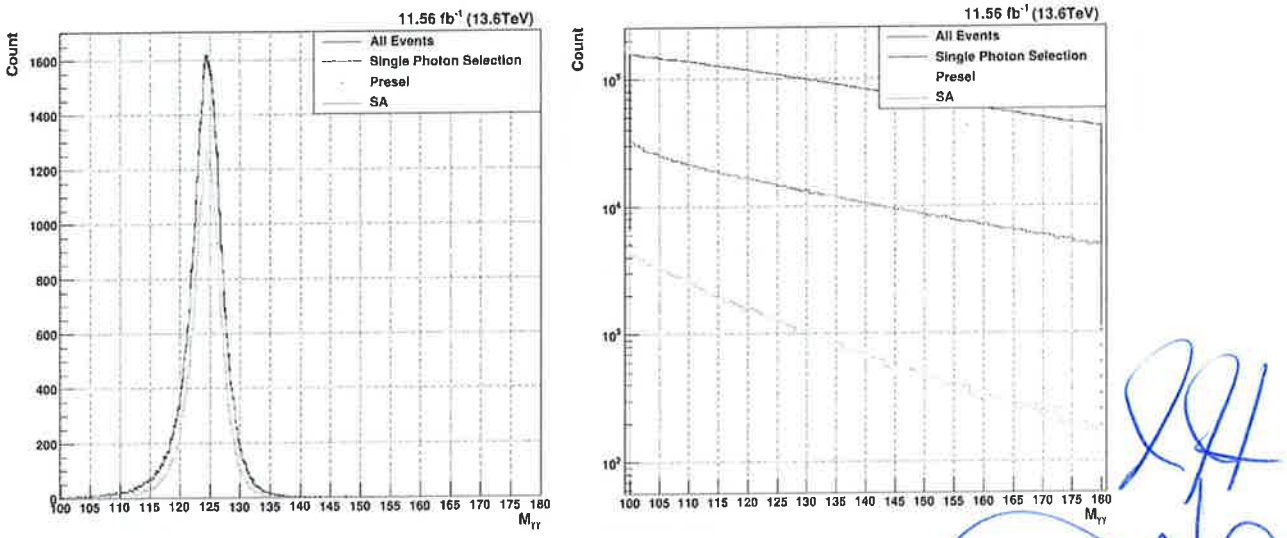


(a) γ + Jet BDT Score

(b) Data/MC BDT Score

Figure 6.16. Plots of the two single photon BDT scores for $H \rightarrow \gamma\gamma$. The γ + Jet scores are on the left, while the data/MC scores are on the right. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

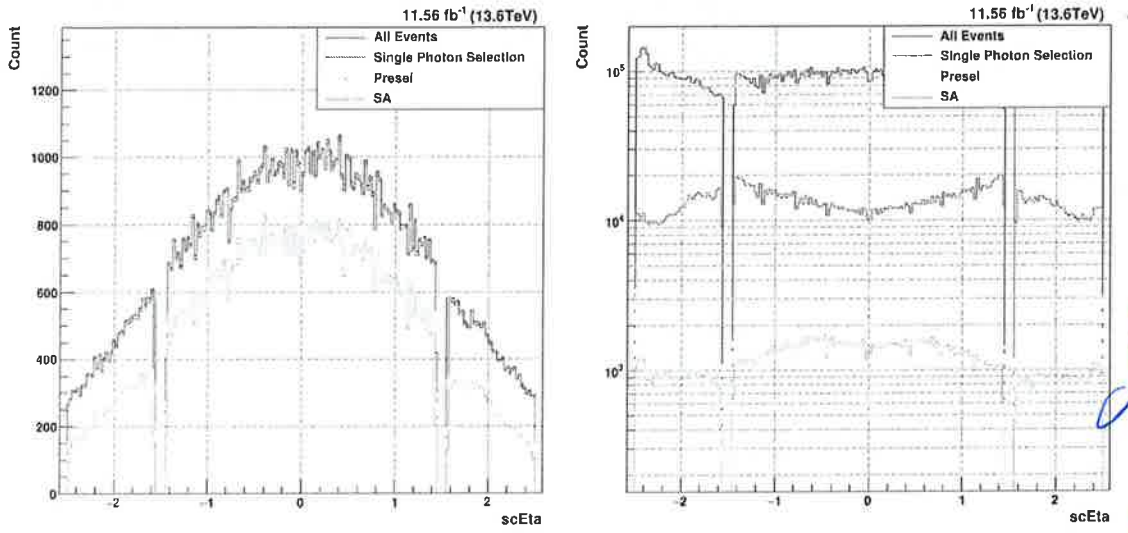
So, then, the aim is to choose a loose set of cuts on these $\gamma + Jet$ and Data/MC BDT scores before applying a tighter final selection on the resulting diphoton combinations. One great advantage of a BDT-based approach is the ease with which these cuts can be adjusted and optimized; while a many-variable preselection has several different cuts to optimize, a binary classifier BDT only has a single output score on which to make cuts. While the goal of this current offline selection is to demonstrate an example analysis that leverages the increased signal efficiency of the MVA-HLT, the existence of these single photon scores allows very easy tuning of the photon preselection for any given analysis. The example cuts chosen below were designed to reduce the signal efficiency by about 2%, which primarily serves to reduce the number of possible photon combinations for final selection; ultimately, the diphoton cuts described in 6.2.4 will be the primary final selection used to achieve the desired data rate.



(a) Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

Figure 6.17. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$

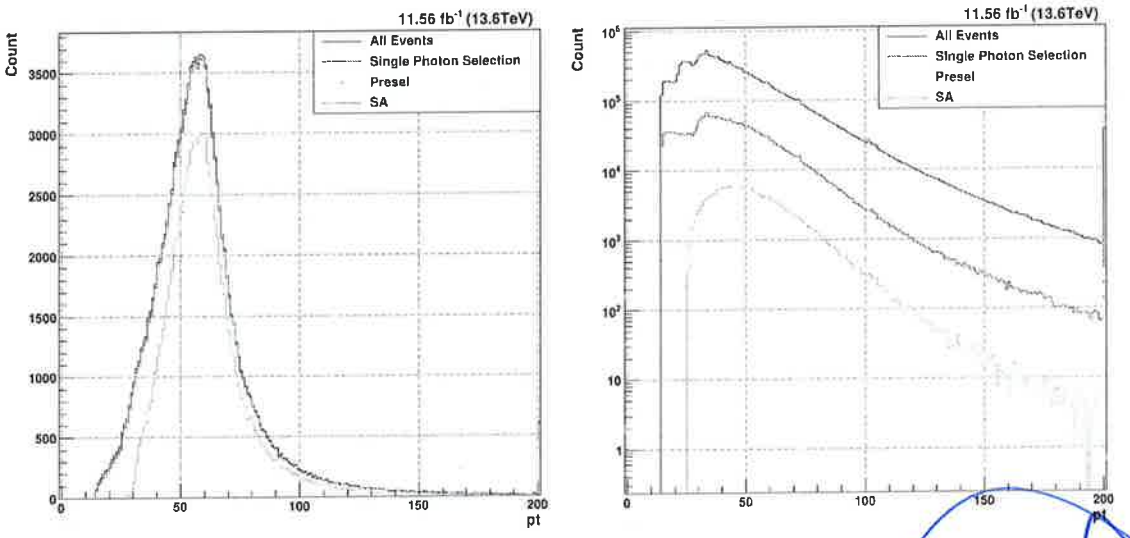
Clearly, as seen in figure 6.17, these chosen single photon cuts (in blue) leave the signal MC relatively untouched. The final chosen signal efficiency for this gluon fusion $H \to \gamma\gamma$ is 98.3%, which is much higher than the standard single photon preselection efficiency of 78.5%. Of course, it can be seen that the background rate is much higher – while the standard preselection cuts achieve a background rejection of 96.8%, the loose single photon BDT cuts only achieve an 85.9% rejection.



(a) Signal MC η_{SC} (b) Data η_{SC}

Figure 6.18. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for η_{SC}. Both the lead and sublead photons are plotted together.

Maximizing the kinematic acceptance of the signal events in η_{SC} is a key consideration to any photon selection. In figure 6.18, the acceptance of the different single photon selections is shown. Given that the signal acceptance is so high for these loose single photon cuts, it can be seen immediately that there are no regions of the detector shown for which the acceptance is problematic. Considering the preselection (yellow), however, a relative drop in efficiency can be seen in the low-η regions at the start of the endcap (near $|\eta| = 1.5$). Investigations showed that the hard R_9 cuts placed on the photons by the standard preselection are responsible for this relative dip in efficiency. Utilizing the R_9 in the BDT, as opposed to these hard cuts, ensures that the kinematic coverage for the loose single photon BDT selection is better than that of the standard preselection.



(a) Signal MC p_T **(b)** Data p_T

Figure 6.19. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for p_T. Both the lead and sublead photons are plotted together.

A major advantage to these loose single photon cuts can be seen in figure 6.19. Clearly, the standard preselection cuts considerably reduce the signal MC kinematic phase space by making relatively tight cuts of $p_T^{Lead} > 35GeV/c^2$ and $p_T^{Sub} > 25GeV/c^2$. While this also leads to a considerable reduction in data rate, the loss of good Higgs events with low-p_T photons may impact future analyses with different kinematics. By comparison, the loose single photon cuts, being based entirely on the single photon BDT scores, do not noticably impact the p_T distributions of either the signal or background.

6.2.4 Creation of Diphoton BDT for $H \to \gamma\gamma$

With the single photon preselection cuts determined, an additional selection must be placed on the diphoton objects. The most obvious continuation of the current strategy is to create an additional MVA that now works on diphoton quantities. Previous analysis strategies have, in fact, employed a diphoton MVA for the classification of events [34]. A new diphoton BDT can be constructed in reference to this, with the aim of making final cuts on this score such that the

Table 6.3. Table of diphoton BDT input variable names, and a short description of each.

Variable Name	Description
$p_T^{Lead}/M_{\gamma\gamma}$ and $p_T^{Sub}/M_{\gamma\gamma}$	The mass-normalized p_T for each photon
η_{sc}^{Lead} and η_{sc}^{Sub}	The supercluster η for each photon
$BDT_{\gamma+jet}^{Lead}$ and $BDT_{\gamma+jet}^{Sub}$	The $\gamma+$ *jet* BDT score for each photon
$BDT_{Data/MC}^{Lead}$ and $BDT_{Data/MC}^{Sub}$	The Data/MC BDT score for each photon
Lead and Sublead Electron Veto	A boolean to reject photons origininating from electrons
$cos(\Delta\phi)$	The cosine of the ϕ separation of the two photons
$\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$	A relative mass resolution estimate

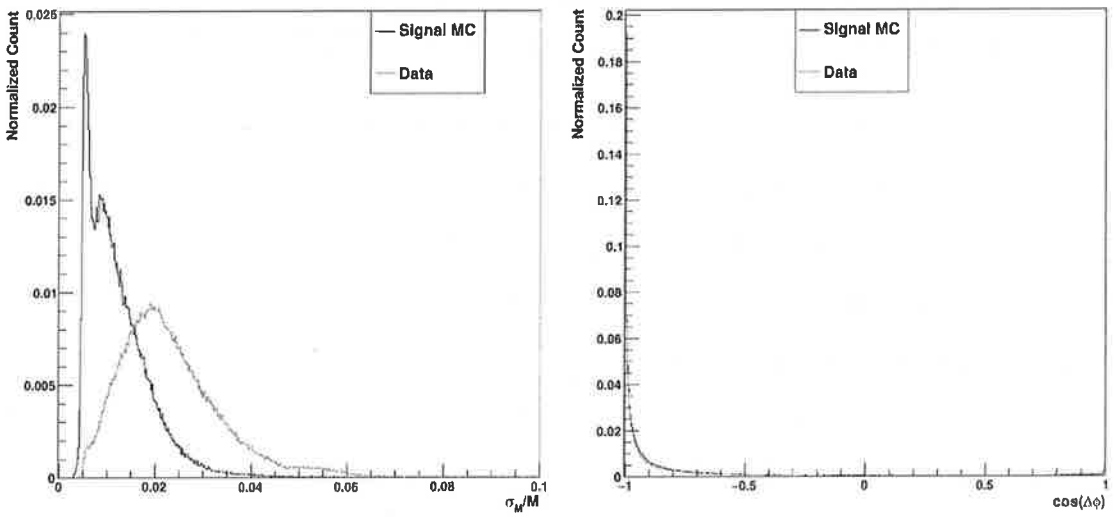
background rate is controlled while maintaining maximum signal efficiency.

While the goal of the single photon BDTs is to reject individual candidates that appear to not be photons, the diphoton BDT aims to properly differentiate diphton candidates coming from $H \rightarrow \gamma\gamma$ from both events that do not contain two real photons and events with two real photons originating from background processes. As a result, the inputs to this model will consist primarily of lead and sublead single photon variables. By providing both the lead and sublead values to one model, correlations can be found that ensure that selected events contain two real photons with the necessary kinematics to possibly comprise a signal event. Below, in table 6.3, each of the input variables is listed and described. These were largely adopted from the Run2 Diphoton photonID, with a few differences.

Since the goal of this model is to differentiate signal diphoton combinations from those in the background, signal MC and data can again be used for the signal and background for the BDT, as with the data/MC BDT described above. Again, the input variable distributions can be examined to see the separation between signal and background class events.

The first two variables are related directly to the kinematics of the diphoton object, and therefore are not split into lead and sublead values. The first, $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, provides a relative invariant mass resolution estimate of the diphoton combination. It can clearly be seen in figure 6.20a, that there is great separation between the signal MC and data events. The signal MC tends to have a much lower $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$, which corresponds to a higher mass resolution. This

is to be expected, as the two selected photons in the $H \to \gamma\gamma$ are much more likely to come from a resonant process than those in the data. The $\cos(\Delta\phi)$, however, does not exhibit the same strong separation. This variable aims to measure the separation in ϕ of the two selected photon candidates. Based on the other selections at play, however, it seems that this variable will not be a strong discrimination variable.



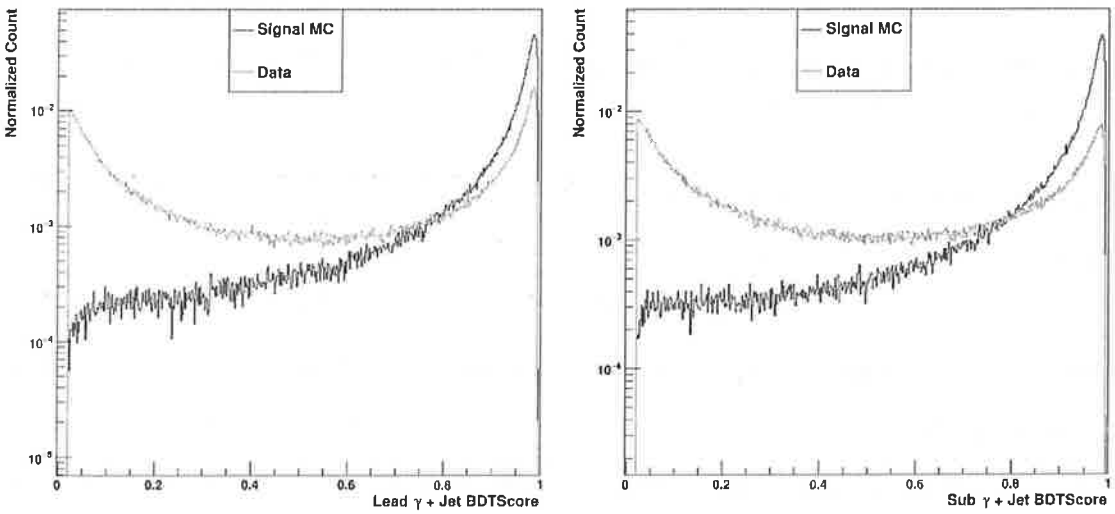
(a) $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ **(b)** $\cos(\Delta\phi)$

Figure 6.20. The two per-event diphoton input variables, $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ and $\cos(\Delta\phi)$. The signal MC is in black, whereas the data events are in red.

The next two pairs of input variables are the single photon BDT scores. Firstly, the γ + Jet single photon BDT scores are shown below in figure 6.21. As with all XGBoost BDT binary classification scores, a value near 0 shows that the algorithm has identified the photon as being part of the background class, while a score near 1 corresponds to an event identified as part of the signal class. For the signal MC, it can be seen that the scores peak very well near 1. This is to be expected, as the signal MC photons here are matched to a generated photon and therefore guaranteed to be a real photon. This effect is clear for both the lead and sublead photons.

Clearly, however, the background photons in red are not constrained entirely near 0. In fact, there are two peaks in both the lead and sublead score distributions. While many of the

background photon candidates can easily be assigned a score near 0, there are also a large number for which the γ + Jet scores are not adequate to differentiate the background from the signal. This likely due, at least in part, to the composition of the data overall. While many of the data events consist of dijet QCD background, there are likely many events for which one or even two real photons have been selected. This is exacerbated by the triggers required for this training sample; here, since either the MVA-HLT trigger or the standard diphoton trigger must be passed to enter the training sample, the odds of a data event containing one or two real photons is much higher than it would be for other data samples. This demonstrate why the γ + Jet BDT alone is not adequate to correctly differentiate between signal and background events.

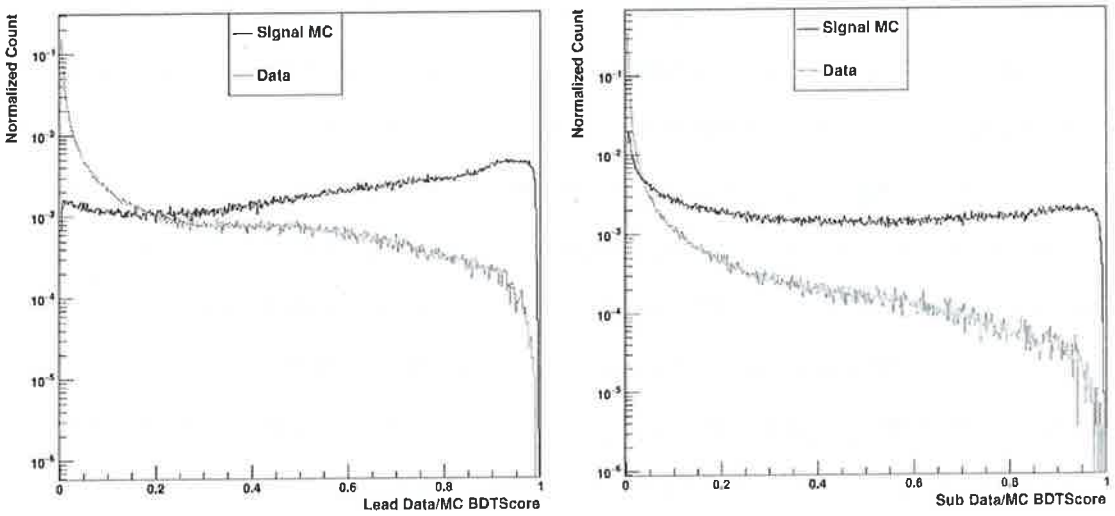


(a) Lead γ + Jet BDT Score

(b) Sub γ + Jet BDT Score

Figure 6.21. The lead and sublead γ + Jet XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

Secondly, the data/MC scores for both the lead and sublead photons are shown in figure 6.22 on a logarithmic scale. As expected, there is clear separation between the signal MC in black and the data in red. It is expected that the signal MC distributions will largely cluster near 1, which corresponds to a positive identification as signal. While this is true for both the lead and sublead scores, the identification of the background is perhaps a stronger effect. It can be seen



(a) Lead Data/MC BDT Score

(b) Sub Data/MC BDT Score

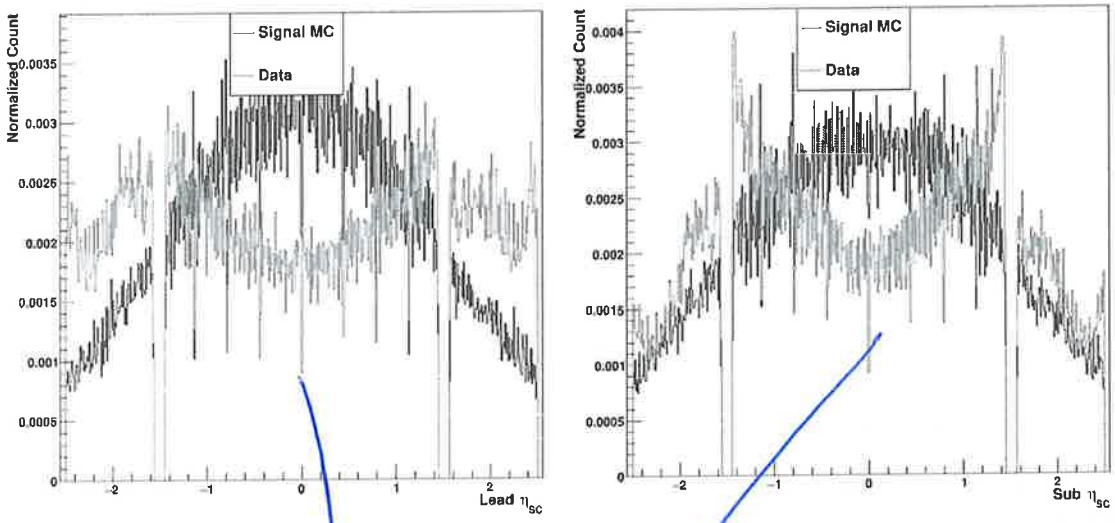
Figure 6.22. The lead and sublead Data/MC XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

that the data events are very closely clustered near 0. Very few of the background events, on the other hand, are placed near a score of 1.

Similarly to the γ + Jet single photon BDT, then, it is clear that this data/MC BDT is not adequate alone to differentiate signal and background. The diphoton BDT, then, is a crucial step of the signal and background selection; by having access to both the single photon BDTs, the diphoton BDT can construct correlations that allow the final model to be considerably more discerning than input scores it uses. As will be shown in chapter 7, this works to great effect.

While the single photon quality scores given by the single photon BDTs are quite helpful, the diphoton kinematics are additionally a very strong source of discrimination power. As sean for both the lead (figure 6.23a) and sublead (figure 6.23a) η_{SC} distributions, the shape of the single Higgs signal MC in black and the data in red are quite different. Due to the kinematics of the Higgs production, as discussed in 1, the signal MC tends to be much more central in the barrel detector (closer to perpendicular to the beamline). Meanwhile, in the endcaps, the amount of signal photons rapidly drops as the edges of the chose η_{SC} range are approached. Due to this

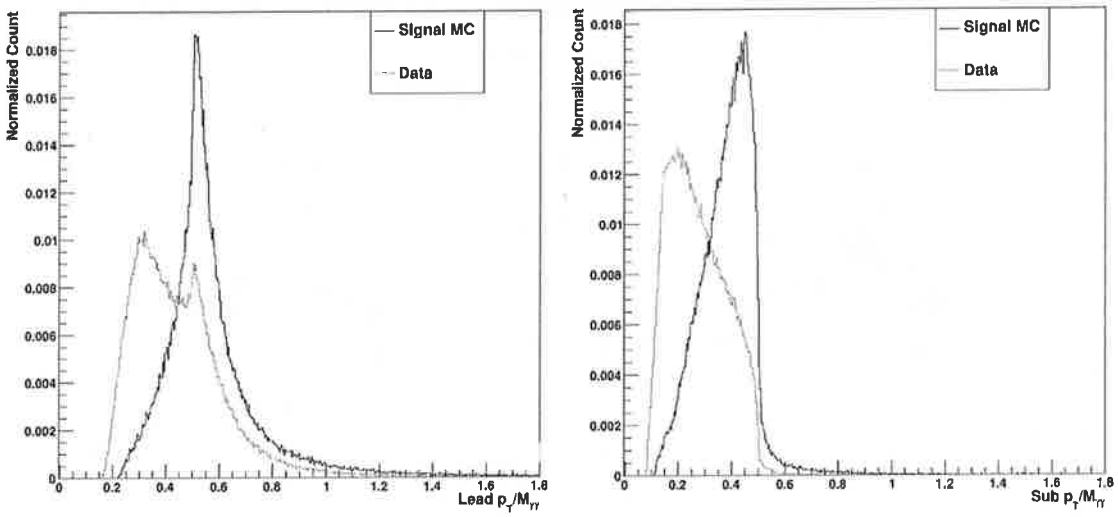
good separation between signal and background, the η_{SC} of the lead and sublead photon provide good discrimination power to the binary classifier BDT.



(a) Lead η_{SC} **(b)** Sub η_{SC}

Figure 6.23. The lead and sublead η_{SC}. The signal MC is in black, whereas the data events are in red.

The other diphoton kinematic quantities used by the diphoton BDT are the $p_T/M_{\gamma\gamma}$ of the two selected photons. This mass-normalized measure of the photon candidates' transverse momentum is used in the standard diphoton preselection given its great discrimination power. Of course, if hard cuts on a variable are successful in separating signal from background, a BDT should be able to leverage the information to even greater effect. Furthermore, as with the p_T cuts, removing the hard cuts on this variable considerably improve the kinematic coverage of the offline selection. In figure 6.24, a clear separation between the signal and background events can be seen for both the lead and sublead photon candidates. Paired with the corresponding η_{SC} values, this lends considerable kinematic discrimination power to the diphoton BDT.



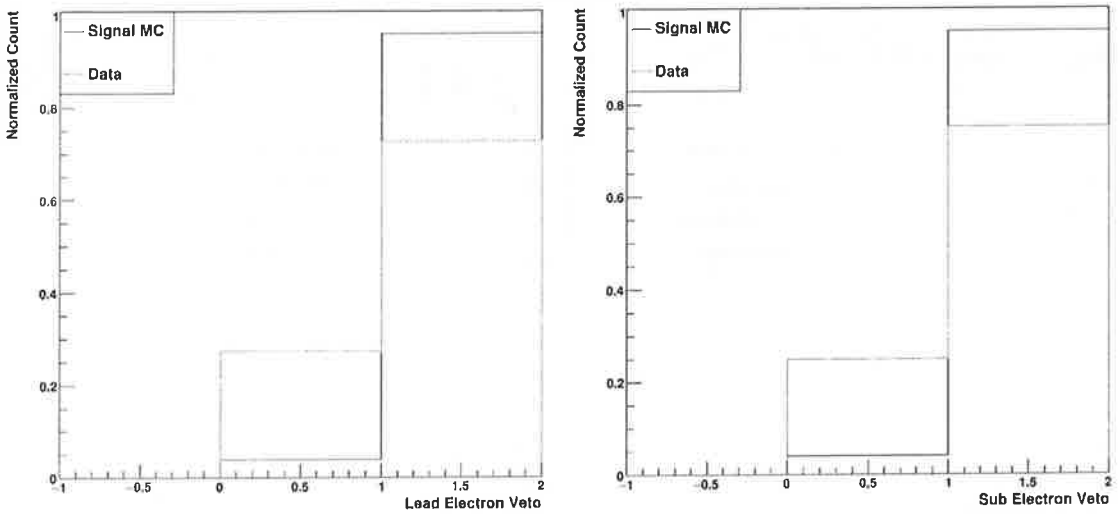
(a) Lead $p_T/M_{\gamma\gamma}$ **(b)** Sub $p_T/M_{\gamma\gamma}$

Figure 6.24. The lead and sublead $p_T/M_{\gamma\gamma}$. The signal MC is in black, whereas the data events are in red.

Finally, the electron veto values can be used to help differentiate between photons and electrons in the final selection. As discussed in section 3.3.1, photons and electrons are quite hard to differentiate purely from ECal information. The electron veto was designed to leverage charged track information to provide a conversion-safe way to reject electrons from photon candidates [24]. This quantity has not historically been deployed in the diphotonMVA; since the standard preselection generally makes a hard requirement on the electron veto, there is no need to use it in the diphotonMVA. By adding the lead and sublead electron veto decision to the diphoton BDT, however, the information can be correlated with the other variables provided to more elegantly reject electrons in the data while still maximizing the signal efficiency.

The values of the electron veto are shown in figure 6.25. Since this is just a veto decision, the variable is saved as a boolean. As a result, the plots show a value of 0 for false (failing the electron veto) and 1 for true (passing the electron veto). A very clear separation can be seen between the signal MC (black) and data (red). Very few of the selected signal MC lead or sublead photon candidates fail the electron veto, wheareas there are a considerable portion of the data

events that fail. It can be seen, however, that there are still many data events that pass the electron veto.



(a) Lead Electron Veto

(b) Sub Electron Veto

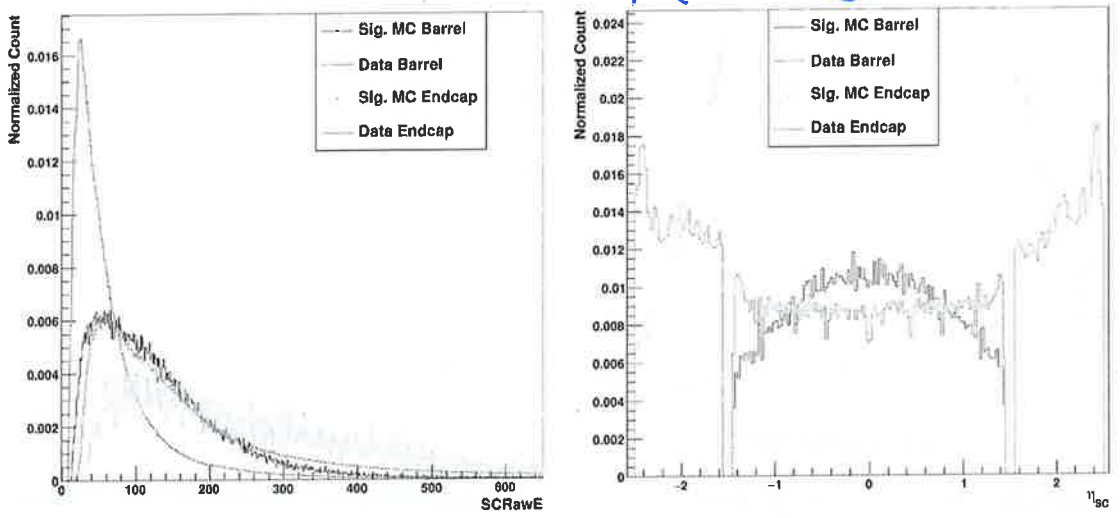
Figure 6.25. The lead and sublead electron veto. The signal MC is in black, whereas the data events are in red.

As with the other two BDTs, the hyperparameters must of course be optimized once the input variables are selected. This was, again, a relatively experimental process. Given the reduced number of input variables relative to the single photon BDTs, lower maximum depths were explored. A max depth of 10 was selected in conjunction with a learning rate of 0.1. This lower max depth and higher learning rate allows to model to still leverage the reduced number of input variables without considerable overtraining.

6.2.5 Creation of a Data/MC BDT for $HH \to b\bar{b}\gamma\gamma$

Of course, since the Data/MC and diphoton BDTs are trained using signal MC and data, they should be retrained for samples with markedly different input features. As discussed in chapter 1, the kinematics of the $HH \to b\bar{b}\gamma\gamma$ differ considerably from the $H \to \gamma\gamma$. Therefore, the same process of training and evaluation must be repeated for these new samples. While

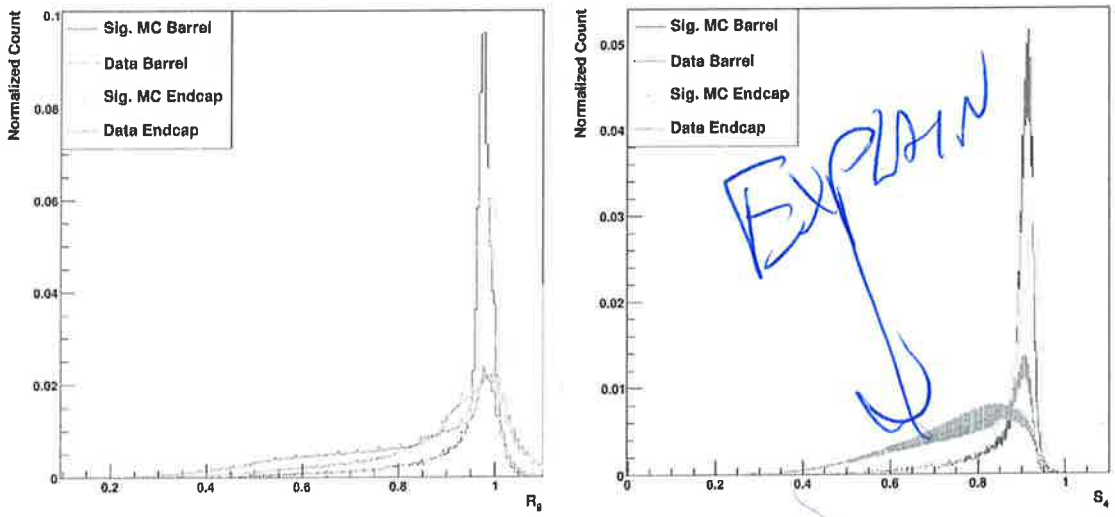
the same data samples are used for this updated analysis, the $HH \rightarrow b\bar{b}\gamma\gamma$ workflow applies additional cuts to the $b\bar{b}$ object. The same input variable distributions for the Data/MC BDT are displayed in figures 6.26-6.31 for these new samples. Major differences between the single and DiHiggs samples will be noted.



(a) Raw Energy **(b)** η_{SC}

Figure 6.26. Plots of the Supercluster raw energy and ηSC. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

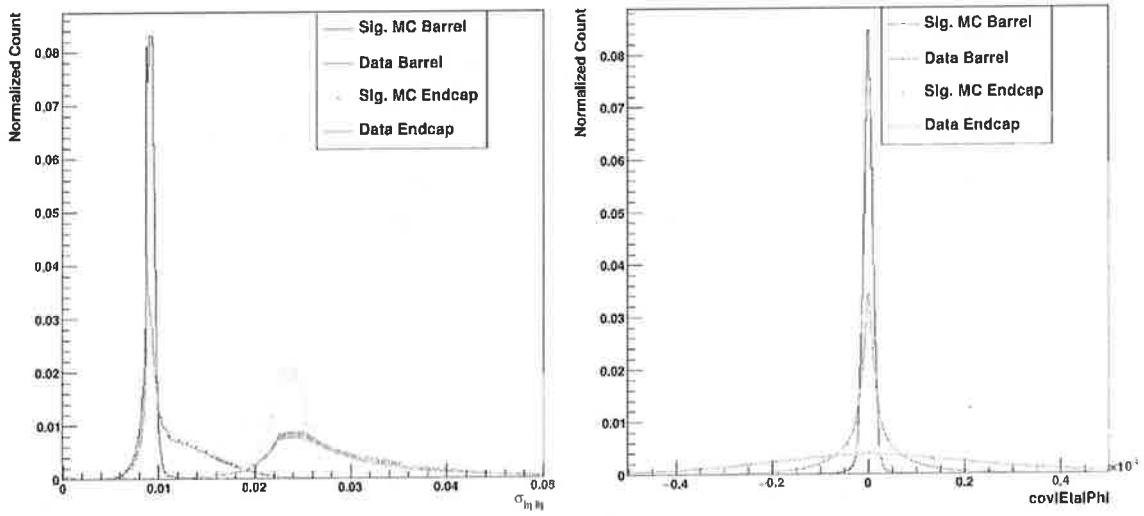
A major difference in η_{SC} exists relative to the single H, as seen in figure 6.26. While the signal MC is relatively similar in the barrel, the endcap is far more concentrated at lower-η values. The data, on the other hand, is considerably more flat in both the endcap and barrel due to the changed kinematics based on the requirements placed on finding a $b\bar{b}$ pair. On the whole, this leads the η_{SC} being used more strongly by the DiHiggs Data/MC BDT. Since less of the MC is in the extreme regions of the endcap, it will be shown that the selection is biased away from selecting data in these regions.



(a) R_9 (b) S_4

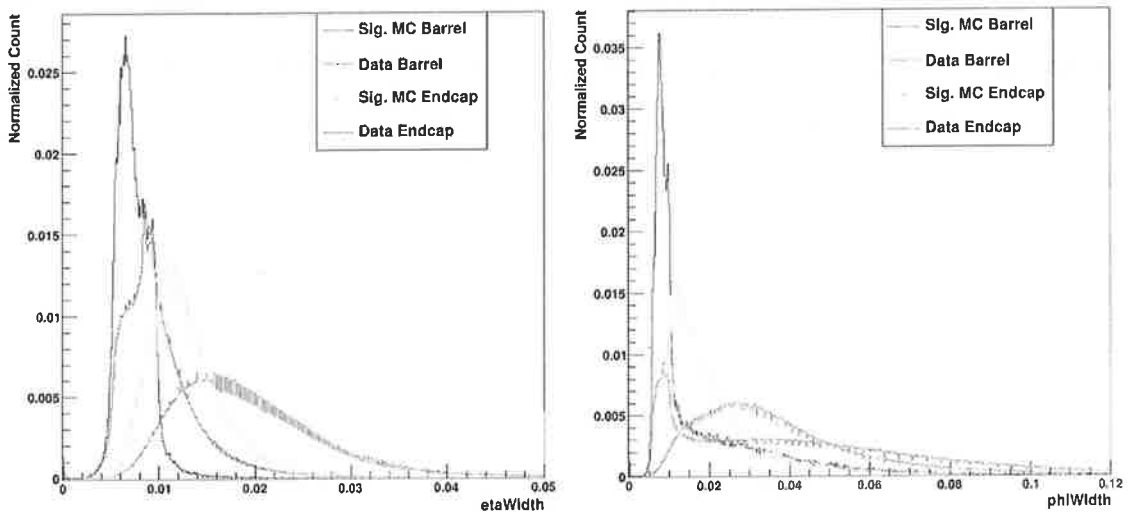
Figure 6.27. Plots of the R_9 and S_4. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

Relative to the data selected by the single Higgs workflow, the DiHiggs data selection has considerably lower values on average of both the R_9 and S_4. This leads to a clearer separation between the signal and background for both the barrel and endcap events.



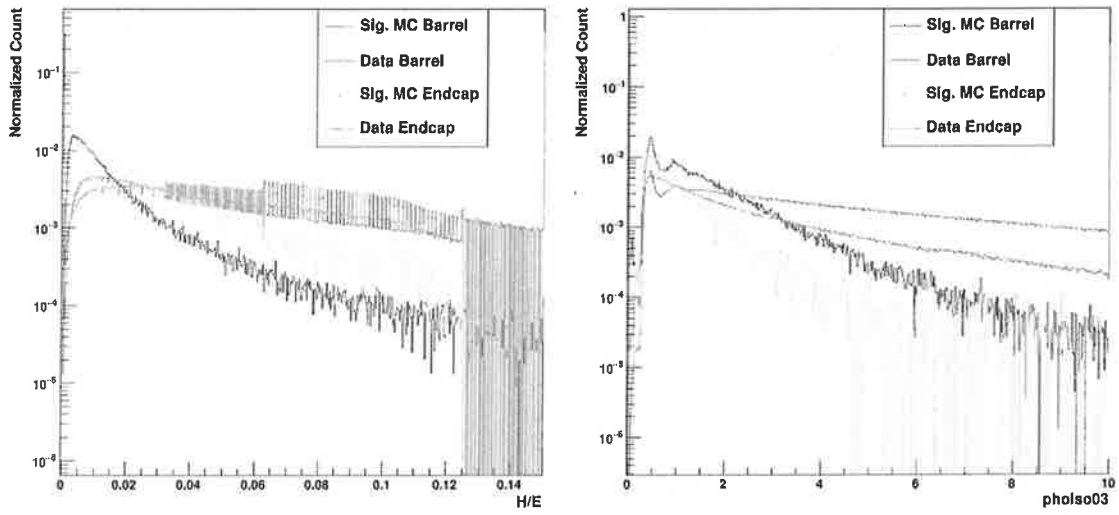
(a) $\sigma_{i\eta i\eta}$ (b) $\mathrm{cov}(i\eta i\phi)$

Figure 6.28. Plots of the $\sigma_{i\eta i\eta}$ and $\mathrm{cov}(i\eta i\phi)$. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.



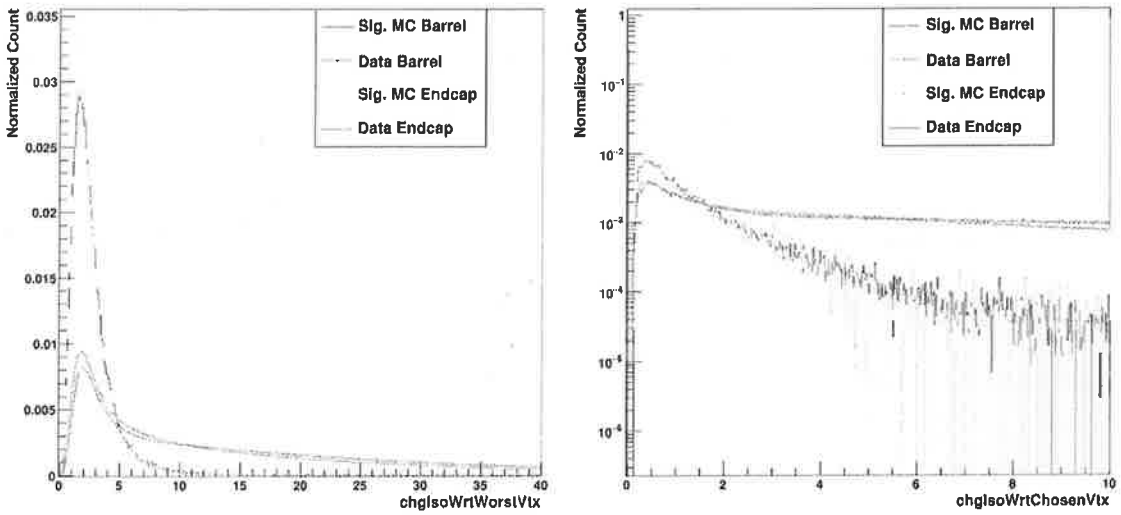
(a) η Width (b) ϕ Width

Figure 6.29. Plots of the η and ϕ widths. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.



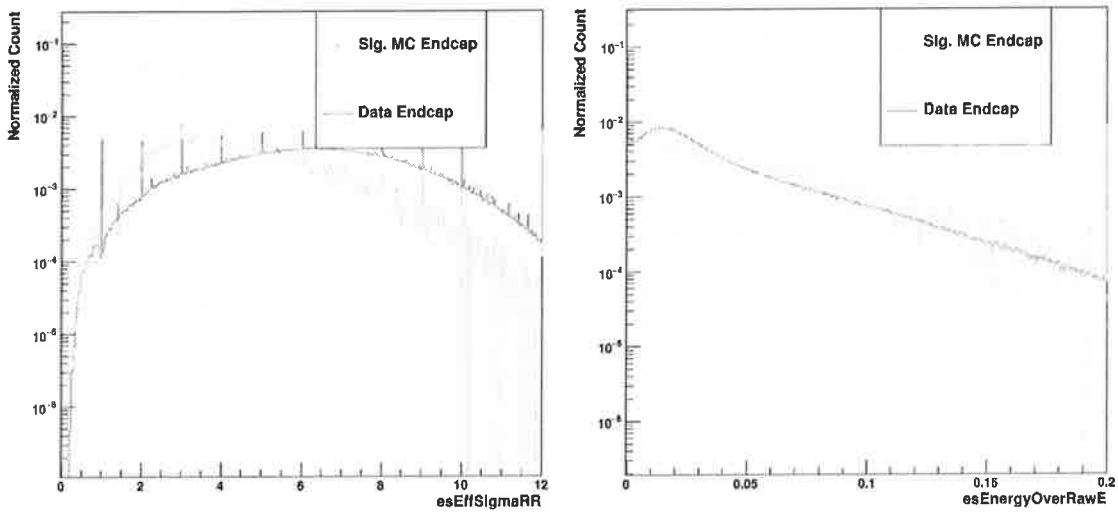
(a) H/E (b) Photon Isolation

Figure 6.30. Plots of the H/E and photon isolation. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.



(a) Chg. Iso. W.r.t. Worst Vertex (b) Chg. Iso. W.r.t. Chosen Vertex

Figure 6.31. Plots of the charged isolations with respect to the worst and chosen vertices. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

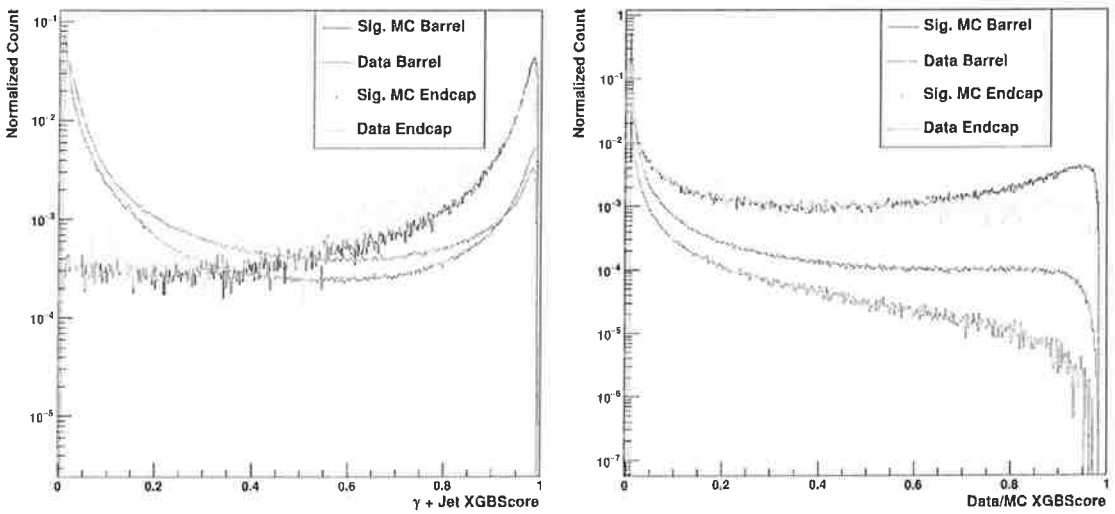


(a) ES Eff. Sigma RR **(b)** ES Energy over Raw E

Figure 6.32. Plots of the ES Eff. Sigma RR and ES Energy over Raw E. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

6.2.6 Determination of Loose Single Photon MVA Cuts for $HH \to b\bar{b}\gamma\gamma$

The resulting single photon BDT score distributions can be seen for $HH \to b\bar{b}\gamma\gamma$ in figure 6.33, where the γ + Jet score can be seen at left and the Data/MC score can be seen at right. The overall features of the distributions are similar to those of the single Higgs, but more exaggerated; the data peaks more strongly at 0 for both the γ + Jet and Data/MC models, while more of the signal MC can be seen near 1 in both cases.

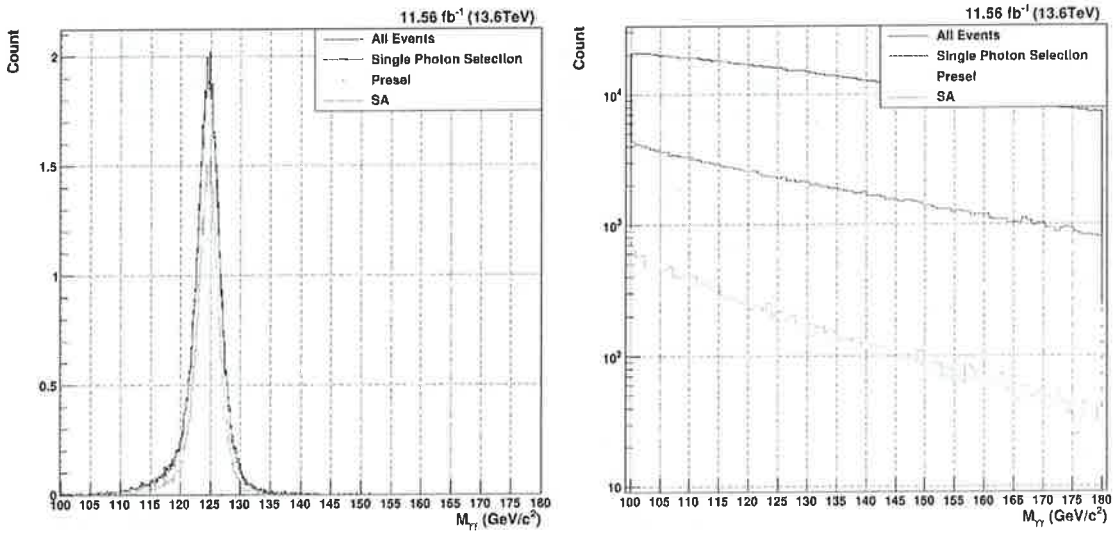


(**a**) γ + Jet BDT Score

(**b**) Data/MC BDT Score

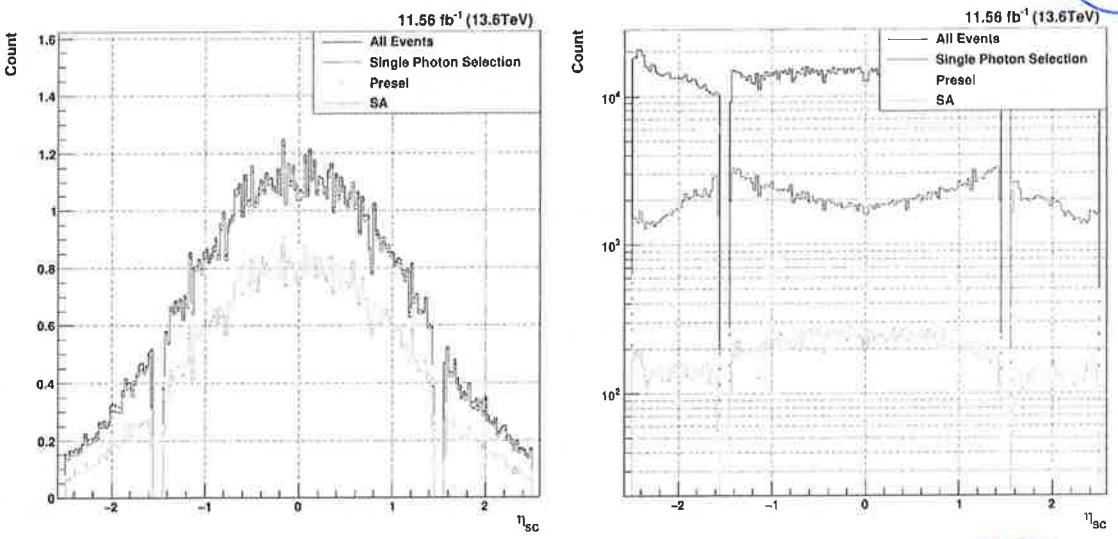
Figure 6.33. Plots of the two single photon BDT scores for $HH \rightarrow b\bar{b}\gamma\gamma$. The γ + Jet scores are on the right, while the data/MC scores are on the left. The signal MC is shown in dark blue for the barrel and light blue for the endcap, whereas the data (background) is shown in red for the barrel and pink for the endcap.

Given the difference in the resulting output BDT score distributions, the loose single photon cuts must be reexamined for the DiHiggs sample. As previously mentioned, this readjustment of the cuts is a relatively straightforward process. Again, the general goal was to achieve an approximately 98% signal efficiency. Relative to the single H cuts, both the γ + Jet and Data/MC cuts are numerically tighter. The chosen cuts yielded a total signal efficiency of 97.35% and a corresponding data acceptance of 14.7%. While both of these values are fractionally lower than that of the single Higgs, there is still ample signal efficiency with which to realize the gains from the trigger level. On the other hand, the preselection yields a signal efficiency of 67.78%, which is considerably lower than that of the single Higgs.



(a) Signal MC $M_{\gamma\gamma}$ **(b)** Data $M_{\gamma\gamma}$

Figure 6.34. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for $M_{\gamma\gamma}$



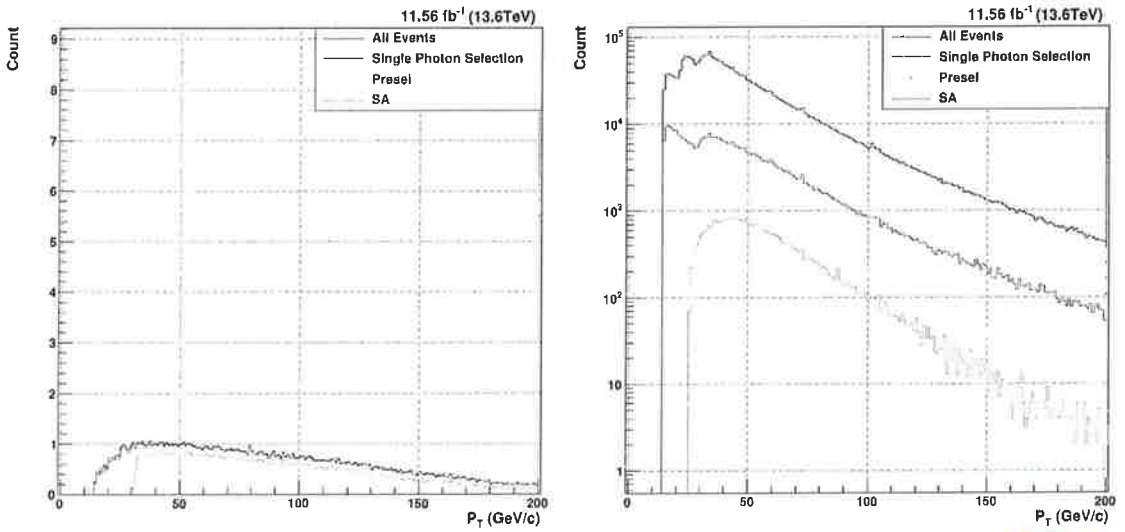
(a) Signal MC η_{SC} **(b)** Data η_{SC}

Figure 6.35. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for η_{SC}

Relative to the singleH selection, the DiHiggs data acceptance is relatively unchanged.

6.34 - 6.36 NOT cited?!

Of course, given the very high signal efficiency of this selection, there is little effect on the signal acceptance. A relevant kinematic difference can be seen, however; the DiHiggs signal MC tends to be far more central in the barrel, with very little extending into the endcap. This highlights, again, why retraining both the single photon Data/MC BDT and the diphoton BDT is so crucial for this sample.



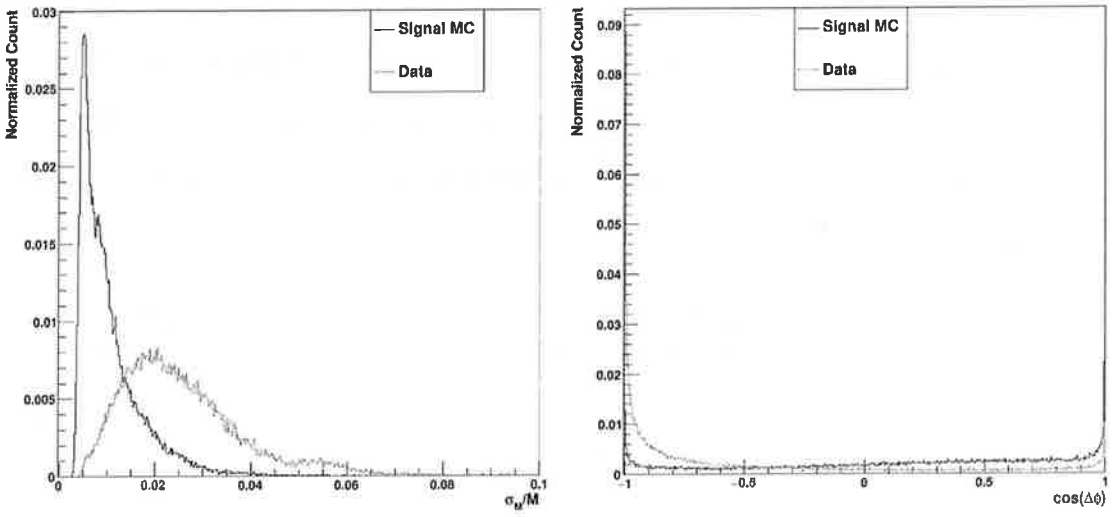
(a) Signal MC p_T (b) Data p_T

Figure 6.36. Acceptance of the MVA-HLT (blue) vs. the current standard diphoton HLT (green) for p_T

The scale of the accepted p_T values is quite skewed by the very large bin content of the overflow bin (above 200 GeV/c). This points to another important kinematic difference – given the more central photon distribution in η_{SC}, the photon candidates tend to have higher p_T values on average.

6.2.7 Creation of a Diphoton BDT for $HH \to b\bar{b}\gamma\gamma$

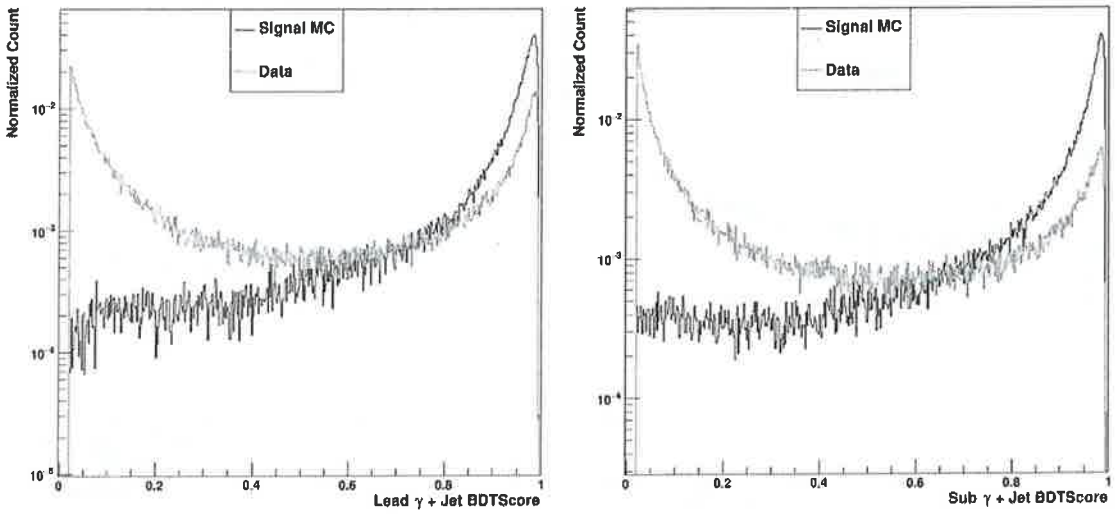
The final step is to consider the recreation of the diphoton BDT for the DiHiggs process. It will be shown in figures 6.37-6.42 that both the kinematic variables and the input BDT score distributions are significantly different relative to the single Higgs.



(a) $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ **(b)** $\cos(\Delta\phi)$

Figure 6.37. The two per-event diphoton input variables,$\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ and $\cos(\Delta\phi)$. The signal MC is in black, whereas the data events are in red.

While the $\sigma(M_{\gamma\gamma})/M_{\gamma\gamma}$ disributions are relatively similar to their single Higgs counterparts, there are minor differences in the $\cos(\Delta\phi)$. The single Higgs for both the data and signal MC was strongly peaked at -1, with a very similar shape for both samples. In the DiHiggs, however, there is far better separation between the signal and background; the data is still relatively well clustered at -1, but the signal MC has a sloped shape favoring higher values (especially increasing approaching 1). This, again, is due to the kinematics of the DiHiggs production. $\text{Cos}(\Delta\phi)$ = -1 corresponds to a $\Delta\phi = \pi$, whereas $\text{Cos}(\Delta\phi)$ = 1 corresponds to a $\Delta\phi = 0$. As expected, then, the DiHiggs signal is more likely to have a lower separation in ϕ. So, the discrimination power of this variable should have increased relative to the single Higgs.

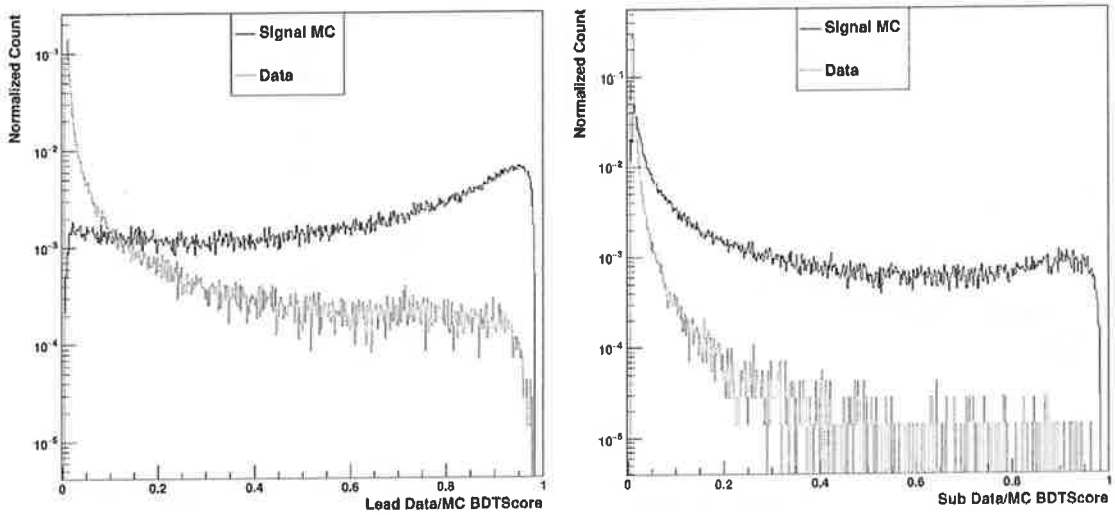


(a) Lead γ + Jet BDT Score

(b) Sub γ + Jet BDT Score

Figure 6.38. The lead and sublead γ + Jet XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

Next, there are clear differences in the γ + Jet BDT score distributions for the DiHiggs. The signal MC, surprisingly, has a relatively similar distribution in scores. The score peaks strongly at 1 for the signal in both the lead and sublead photon candidate. The data, however, has moved closer to 0. Especially in the sublead photon candidate, the peak near 0 is taller than the peak near 1. In the single Higgs, the two peaks were of a more similar size. Even though this γ + Jet BDT was mot retrained, it appears to perform better on separating DiHiggs signal MC from data.

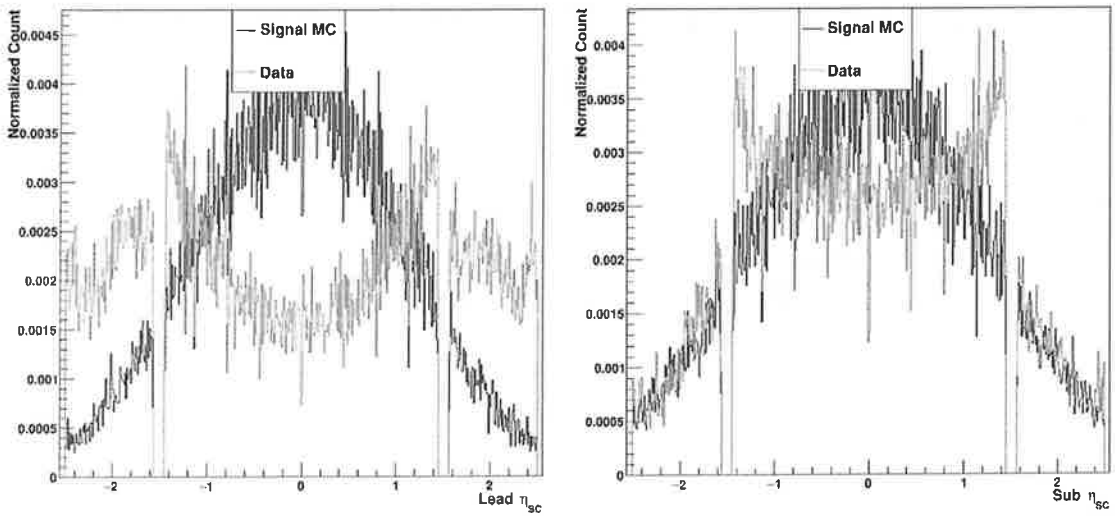


(a) Lead Data/MC BDT Score

(b) Sub Data/MC BDT Score

Figure 6.39. The lead and sublead Data/MC XGBoost BDT scores. The signal MC is in black, whereas the data events are in red.

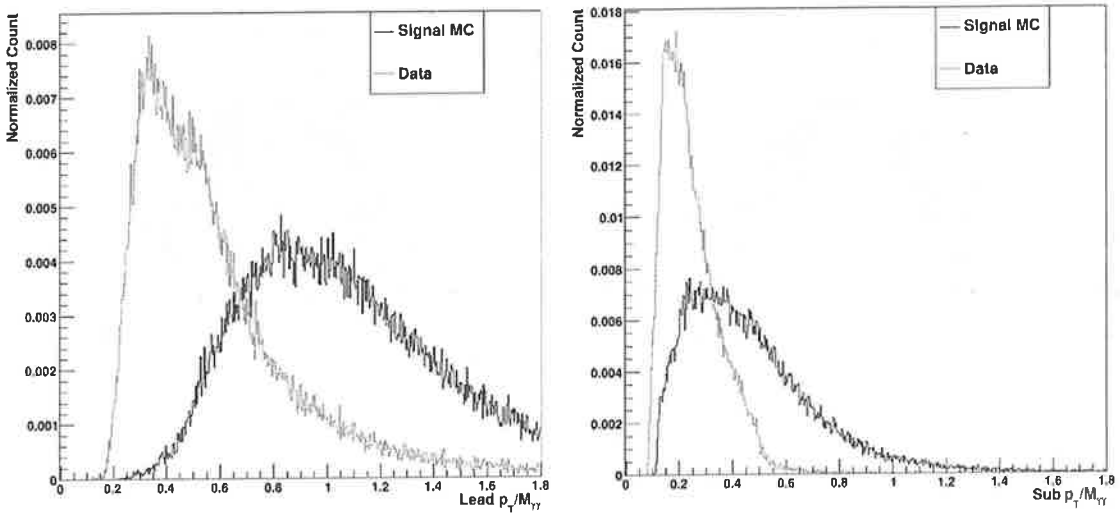
The single photon Data/MC BDT scores are quite well-separated for the lead photon, perhaps moreso than they were for the single Higgs sample. Clearly, as well, the sublead scores for the data are very well-placed near 0. Again, in conjunction with the γ + Jet scores shown in figure 6.38, these data/MC BDT scores will have strong discrimination power.



(a) Lead η_{SC} **(b)** Sub η_{SC}

Figure 6.40. The lead and sublead η_{SC}. The signal MC is in black, whereas the data events are in red.

As mentioned for the selection of the single photon cuts, the η_{SC} distributions for the diHiggs samples are more separated than those of the single Higgs. Especially in the lead photon candidate, there is very little data in the center of the barrel, while the signal MC sees its strongest peak in this region. The endcap shapes are rather different for the lead photon, but much more similar for the sublead photon. Again, due to the difference in kinematics relative to the single Higgs, these differences are to be expected.



(a) Lead $p_T/M_{\gamma\gamma}$ **(b)** Sub $p_T/M_{\gamma\gamma}$

Figure 6.41. The lead and sublead $p_T/M_{\gamma\gamma}$. The signal MC is in black, whereas the data events are in red.

While the separation between the signal and background for both lead and sublead $p_T/M_{\gamma\gamma}$ is still quite good (as seen in figure 6.41), the distributions for the signal MC are quite different than those for the single Higgs. Specifically, the values tend to be, on average, a bit higher for both the lead and sublead. Additionally, the spread of the values for the signal MC is quite a bit higher, which likely reduces the discrimination power of the $p_T/M_{\gamma\gamma}$ for the diHiggs samples.